

We next show how this procedure may be reversed: Given a class in $H^2(G, A)$ we construct an extension E_f whose corresponding factor set is in the given class in $H^2(G, A)$. The process generalizes the semidirect product construction of Section 5.5 (which is the special case when f is the zero cocycle representing the trivial class).

Note first that any 2-cocycle arising from the factor set of an extension as above where the section μ is normalized satisfies the condition in (31).

Definition. A 2-cocycle f such that $f(g, 1) = 0 = f(1, g)$ for all $g \in G$ is called a *normalized 2-cocycle*.

The construction of E_f is a little simpler when f is a normalized cocycle and for simplicity we indicate the construction in this case (the minor modifications necessary when f is not normalized are indicated in Exercise 4).

We first see that any 2-cocycle f lies in the same cohomology class as a normalized 2-cocycle. Let $d_1 f_1$ be the 2-coboundary of the constant function f_1 on G whose value is $f(1, 1)$. Then $f(1, 1) = d_1 f_1(1, 1)$, and one easily checks from the 2-cocycle condition that $f - d_1 f_1$ is normalized.

We may therefore assume that our cohomology class in $H^2(G, A)$ is represented by the normalized 2-cocycle f . Let E_f be the set $A \times G$, and define a binary operation on E_f by

$$(a_1, g)(a_2, h) = (a_1 + g \cdot a_2 + f(g, h), gh) \quad (17.34)$$

where, as usual, $g \cdot a_2$ denotes the module action of G on A . It is straightforward to check that the group axioms hold: Since f is normalized, the identity element is $(0, 1)$ and inverses are given by

$$(a, g)^{-1} = (-g^{-1} \cdot a - f(g^{-1}, g), g^{-1}). \quad (17.35)$$

The cocycle condition implies the associative law by calculations similar to (32) and (33) earlier — the details are left as exercises.

Since f is a normalized 2-cocycle, $A^* = \{(a, 1) \mid a \in A\}$ is a subgroup of E_f , and the map $\iota^* : a \mapsto (a, 1)$ is an isomorphism from A to A^* . Moreover, from (34) and (35) it follows that

$$(0, g)(a, 1)(0, g)^{-1} = (g \cdot a, 1) \quad \text{for all } g \in G \text{ and all } a \in A. \quad (17.36)$$

Since E_f is generated by A^* together with the set of elements $(0, g)$ for $g \in G$, (36) implies that A^* is a normal subgroup of E_f . Furthermore, it is immediate from (34) that the map $\pi^* : (a, g) \mapsto g$ is a surjective homomorphism from E_f to G with kernel A^* , i.e., $E_f/A^* \cong G$. Thus

$$1 \longrightarrow A \xrightarrow{\iota^*} E_f \xrightarrow{\pi^*} G \longrightarrow 1 \quad (17.37)$$

is a specific extension of G by A , where (36) ensures also that the action of G on A by conjugation in this extension is the module action specified in determining the 2-cocycle f in $H^2(G, A)$. The extension sequence (37) shows that this extension has the normalized section $\mu(g) = (0, g)$ whose corresponding normalized factor set is f . Note that this proves not only that every cohomology class in $H^2(G, A)$ arises from

some extension E , but that every normalized 2-cocycle arises as the normalized factor set of some extension.

Finally, suppose f' is another normalized 2-cocycle in the same cohomology class in $H^2(G, A)$ as f and let $E_{f'}$ be the corresponding extension. If f and f' differ by the coboundary of $f_1 : G \rightarrow A$ then $f(g, h) - f'(g, h) = gf_1(h) - f_1(gh) + f_1(g)$ for all $g, h \in G$. Setting $g = h = 1$ shows that $f_1(1) = 0$. Define

$$\beta : E_f \longrightarrow E_{f'} \quad \text{by} \quad \beta((a, g)) = (a + f_1(g), g).$$

It is immediate that β is a bijection, and

$$\begin{aligned} \beta((a_1, g)(a_2, h)) &= \beta((a_1 + g \cdot a_2 + f(g, h), gh)) \\ &= (a_1 + g \cdot a_2 + f(g, h) + f_1(gh), gh) \\ &= (a_1 + f_1(g) + g \cdot (a_2 + f_1(h)) + f'(g, h), gh) \\ &= (a_1 + f_1(g), g)(a_2 + f_1(h), h) = \beta((a_1, g))\beta((a_2, h)) \end{aligned}$$

shows that β is an isomorphism from E_f to $E_{f'}$.

The restriction of β to A is given by $\beta((a, 1)) = (a + f_1(1), 1) = (a, 1)$, so β is the identity map on A . Similarly β is the identity map on the second component of (a, g) , so β induces the identity map on the quotient G . It follows that β defines an equivalence between the extensions E_f and $E_{f'}$. This shows that the equivalence class of the extension E_f depends only on the cohomology class of f in $H^2(G, A)$.

We summarize this discussion in the following theorem.

Theorem 36. Let A be a G -module. Then

- (1) A function $f : G \times G \rightarrow A$ is a normalized factor set of some extension E of G by A (with conjugation given by the G -module action on A) if and only if f is a normalized 2-cocycle in $Z^2(G, A)$.
- (2) There is a bijection between the equivalence classes of extensions E as in (1) and the cohomology classes in $H^2(G, A)$. The bijection takes an extension E into the class of a normalized factor set f for E associated to any normalized section μ of G into E , and takes a cohomology class c in $H^2(G, A)$ to the extension E_f defined by the extension (37) for any normalized cocycle f in the class c .
- (3) Under the bijection in (2), split extensions correspond to the trivial cohomology class.

Corollary 37. Every extension of G by the abelian group A splits if and only if $H^2(G, A) = 0$.

Corollary 38. If A is a finite abelian group and $(|A|, |G|) = 1$ then every extension of G by A splits.

Proof: This follows immediately from Corollary 29 in Section 2.

We can use Corollary 38 to prove the same result without the restriction that A be an abelian group.

Theorem 39. (Schur's Theorem) If E is any finite group containing a normal subgroup N whose order and index are relatively prime, then N has a complement in E .

Remark: Recall that a subgroup whose order and index are relatively prime is called a *Hall subgroup*, so Schur's Theorem says that every normal Hall subgroup has a complement that splits the group as a semidirect product.

Proof: We use induction on the order of E . Since we may assume $N \neq 1$, let p be a prime dividing $|N|$ and let P be a Sylow p -subgroup of N . Let E_0 be the normalizer in E of P and let $N_0 = N \cap E_0$. By Frattini's Argument (Proposition 6 in Section 6.1) $E = E_0N$. It follows from the Second Isomorphism Theorem that N_0 is a (normal) Hall subgroup of E_0 and $|E_0 : N_0| = |E : N|$ (cf. Exercise 10 of Section 3.3).

If $E_0 < E$, then by induction applied to N_0 in E_0 we obtain that E_0 contains a complement K to N_0 . Since $|K| = |E_0 : N_0|$, K is also a complement to N in E , as needed. Thus we may assume $E_0 = E$, i.e., P is normal in E .

Since the center of P , $Z(P)$, is characteristic in P , it is normal in E (cf. Section 4.4). If $Z(P) = N$, then N is abelian and the theorem follows from Corollary 38. Thus we may assume $Z(P) \neq N$. Let bars denote passage to the quotient group $E/Z(P)$. Then $\bar{N}$ is a normal Hall subgroup of $\bar{E}$. By induction it has a complement $\bar{K}$ in $\bar{E}$. Let E_1 be the complete preimage of $\bar{K}$ in E . Then $|E_1| = |\bar{K}||Z(P)| = |E/N||Z(P)|$, so $Z(P)$ is a normal Hall subgroup of E_1 . By induction $Z(P)$ has a complement in E_1 which is seen by order considerations to also be a complement to N in E . This completes the proof.

Examples

- (1) If $G = Z_2$ and $A = \mathbb{Z}/2\mathbb{Z}$ then G acts trivially on A and so $H^2(G, A) = A^G/NA = \mathbb{Z}/2\mathbb{Z}$ by the computation of the cohomology of cyclic groups in Section 2, so by Theorem 36 there are precisely two inequivalent extensions of G by A . These are the cyclic group of order 4 and the Klein 4-group, the latter being split and hence corresponding to the trivial class in H^2 .
- (2) If $G = \langle g \rangle \cong Z_2$ and $A = \langle a \rangle \cong \mathbb{Z}/4\mathbb{Z}$ is a group of order 4 on which G acts trivially, then $H^2(G, A) = A/2A \cong \mathbb{Z}/2\mathbb{Z}$ by the computation of the cohomology of cyclic groups. As in the previous example there are two inequivalent extensions of G by A ; evidently these are the groups Z_8 and $Z_4 \times Z_2$, the latter split extension corresponding to the trivial cohomology class.

If $E = \langle r \rangle \times \langle s \rangle$ denotes the split extension of G by A , where $|r| = 4$ and $|s| = 2$, then $\mu_i(g) = r^i s$ for $i = 0, \dots, 3$ give the four normalized sections of G in E . The sections μ_0, μ_2 both give the zero factor set f . The sections μ_1, μ_3 both give the factor set f' with $f'(g, g) = a^2 \in A$. Both f and f' give normalized 2-cocycles lying in the trivial cohomology class of $H^2(G, A)$. The extension E_f corresponding to the zero 2-cocycle f is the group with the elements $(a, 1)$ and $(1, g)$ as the usual generators (of orders 4 and 2, respectively) for $Z_4 \times Z_2$. In $E_{f'}$, however, $(a, 1)$ has order 4 but so does $(1, g)$ since $(1, g)^2 = (f'(g, g), g^2) = (a^2, 1)$. The 2-cocycles f and f' differ by the coboundary f_1 with $f_1(1) = 1$ and $f_1(g) = r$. The isomorphism $\beta(a, g) = (a + f_1(g), g)$ from E_f to $E_{f'}$ maps the generators $(a, 1)$ and $(1, g)$ of E_f to the generators $(a, 1)$ and (a, g) of $E_{f'}$ and gives the explicit equivalence of these two extensions.

The situation where G acts on A by inversion is handled in Exercise 3.

(3) Suppose $G = Z_2$ and A is the Klein 4-group. If G acts nontrivially on A then G interchanges two of the nonidentity elements, say a and b , of A and fixes the third nonidentity element c . Then $A^G = NA = \{1, c\}$ and so $H^2(G, A) = 0$, and so every extension E of G by A splits. This can be seen directly, as follows. Since the action is nontrivial, such a group must be nonabelian, hence must be D_8 . From the lattice of D_8 in Section 2.5 one sees that for each Klein 4-group there is a subgroup of order 2 in D_8 not contained in the 4-group and that subgroup splits the extension.

If G acts trivially on A then $H^2(G, A) = A/2A \cong A$, so there are 4 inequivalent extensions of G by A in this case. These are considered in Exercise 1.

Example: (Groups of Order 8 and $H^2(Z_2 \times Z_2, \mathbb{Z}/2\mathbb{Z})$)

Let $G = \{1, a, b, c\}$ be the Klein 4-group and let $A = \mathbb{Z}/2\mathbb{Z}$. The 2-group G must act trivially on A . The elements of $H^2(G, A)$ classify extensions E of order 8 which has a quotient group by some Z_2 subgroup that is isomorphic to the Klein 4-group. Although there are, up to group isomorphism, only four such groups, we shall see that there are *eight* inequivalent extensions.

Since $G \times G$ has 16 elements, we have $|C^2(G, A)| = 2^{16}$. The cocycle condition (26) here reduces to

$$f(g, h) + f(gh, k) = f(h, k) + f(g, hk) \quad \text{for all } g, h, k \in G. \quad (17.38)$$

The following relations hold for the subgroup $Z^2(G, A)$ of cocycles:

- (1) $f(g, 1) = f(1, g) = f(1, 1)$, for all $g \in G$
- (2) $f(g, 1) + f(g, a) + f(g, b) + f(g, c) = 0$, for all $g \in G$
- (3) $f(1, h) + f(a, h) + f(b, h) + f(c, h) = 0$, for all $h \in G$.

The first of these come from (38) by setting $h = k = 1$ and by setting $g = h = 1$. The other two relations come from (38) by setting $g = h$ and $h = k$, respectively, using relations (1) and (2). It follows that every 2-cocycle f can be represented by a vector $(\alpha, \beta, \gamma, \delta, \epsilon)$ in $\mathbb{F}_2$ where

$$\begin{aligned} \alpha &= f(1, g) = f(g, 1), \text{ for all } g \in G, \\ \beta &= f(a, a), \quad \gamma = f(a, b), \quad \delta = f(b, a), \quad \epsilon = f(b, b) \end{aligned}$$

because the relations above then determine the remaining values of f :

$$\begin{aligned} f(a, c) &= \alpha + \beta + \gamma & f(b, c) &= \alpha + \delta + \epsilon & f(c, a) &= \alpha + \beta + \delta \\ f(c, b) &= \alpha + \gamma + \epsilon & f(c, c) &= \alpha + \beta + \gamma + \epsilon. \end{aligned}$$

It follows that $|Z^2(G, A)| \leq 2^5$. Although one could eventually show that every function satisfying these relations is a 2-cocycle (hence the order is exactly 32), this will follow from other considerations below.

A cocycle f is a coboundary if there is a function $f_1 : G \rightarrow A$ such that

$$f(g, h) = f_1(h) - f_1(gh) + f_1(g), \quad \text{for all } g, h \in G.$$

This coboundary condition is easily seen to be equivalent to the conditions:

- (i) $f(g, 1) = f(1, g) = f(g, g)$ for all $g \in G$, and
- (ii) $f(g, h) = f(g', h')$ whenever g, h are distinct nonidentity elements and so are g', h' .

These relations are equivalent to $\alpha = \beta = \epsilon$ and $\gamma = \delta$. Thus $B^2(G, A)$ consists of the vectors $(\alpha, \alpha, \gamma, \gamma, \alpha)$, and so $H^2(G, A)$ has dimension at most 3 (i.e., order at most $2^3 = 8$). It is easy to see that $\{(0, \beta, \gamma, 0, \epsilon)\}$ with β, γ , and ϵ in $\mathbb{F}_2$ gives a set of representatives for $Z^2(G, A)/B^2(G, A)$, and each of these representative cocycles is normalized. We

now prove $|H^2(G, A)| = 8$ (and also that $|Z^2(G, A)| = 2^5$) by explicitly exhibiting eight inequivalent group extensions,

Suppose E is an extension of G by A , where for simplicity we assume $A \leq E$. If $\mu: G \rightarrow E$ is a section, the factor set for E associated to μ satisfies

$$\mu(g)\mu(h) = f(g, h)\mu(gh).$$

The group E is generated by $\mu(a)$, $\mu(b)$ and A , and A is contained in the center of E since G acts trivially on A . Hence E is abelian if and only if $\mu(a)\mu(b) = \mu(b)\mu(a)$, which by the relation above occurs if and only if $f(a, b) = f(b, a)$. If g is a nonidentity element in G , we also see from the relation above that $\mu(g)$ is an element of order 2 in E if and only if $f(g, g) = 0$. Because A is contained in the center of E , both elements in any nonidentity coset $A\mu(g)$ have the same order (either 2 or 4).

There are four groups of order 8 containing a normal subgroup of order 2 with quotient group isomorphic to the Klein 4-group: $Z_2 \times Z_2 \times Z_2$, $Z_4 \times Z_2$, D_8 , and Q_8 .

The group $E \cong Z_2 \times Z_2 \times Z_2$ is the split extension of G by A and has $f = 0$ as factor set.

When $E \cong Q_8$, in the usual notation for the quaternion group $A = \langle -1 \rangle$. In this (non-abelian) group every nonidentity coset consists of elements of order 4, and this property is unique to Q_8 , so the resulting factor set f satisfies $f(g, g) \neq 0$ for all nonidentity elements in G .

When $E \cong Z_4 \times Z_2 = \langle x \rangle \times \langle y \rangle$ we must have $A = \langle x^2 \rangle$. The cosets Ax and Axy both consist of elements of order 4, and the coset Ay consists of elements of order 2, so exactly one of $\mu(a)$, $\mu(b)$ or $\mu(c)$ is an element of order 2 and the other two must be of order 4. This suggests three homomorphisms from E to G , defined on generators by

$$\begin{aligned}\pi_1(y) &= a & \pi_1(x) &= b \\ \pi_2(y) &= b & \pi_2(x) &= a. \\ \pi_3(y) &= c & \pi_3(x) &= a\end{aligned}$$

Each of these homomorphisms maps surjectively onto G , has A as kernel, and has $\mu(a)$ (respectively, $\mu(b)$, $\mu(c)$) an element of order 2 in E . Any isomorphism of E with itself that is the identity on A must take the unique nonidentity coset Ay of A consisting of elements of order 2 to itself. Hence any extension equivalent to the extension E_1 defined by π_1 also maps y to a (since the equivalence is the identity on G). It follows that the three extensions defined by π_1 , π_2 and π_3 are inequivalent.

The situation when $E \cong D_8 = \langle r, s \rangle$ is similar. In this case $A = \langle r^2 \rangle$, the cosets As and Asr consist of elements of order 2, and the coset Ar consists of elements of order 4. In this case exactly one of $\mu(a)$, $\mu(b)$ or $\mu(c)$ is an element of order 4 and the other two are of order 2, suggesting the three homomorphisms defined on generators by

$$\begin{aligned}\pi_1(r) &= a & \pi_1(s) &= b \\ \pi_2(r) &= b & \pi_2(s) &= a. \\ \pi_3(r) &= c & \pi_3(s) &= a\end{aligned}$$

As before, the corresponding extensions are inequivalent.

The existence of 8 inequivalent extensions of G by A proves that $|H^2(G, A)| = 8$, and hence that these are a complete list of all the inequivalent extensions. In particular, the extension $E'_1 \cong Z_4 \times Z_2$ defined by the homomorphism π'_1 mapping y to a and x to c must be equivalent to the extension E_1 above (and similarly for the other two extensions isomorphic to $Z_4 \times Z_2$ and the three extensions for D_8). This proves the existence of certain outer automorphisms for these groups, cf. Exercise 9.

Remark: For any prime p the cohomology groups of the elementary abelian group E_{p^m} with coefficients in the finite field $\mathbb{F}_p$ may be determined by relating them to the cohomology groups of the factors in the direct product as mentioned at the end of Section 2. In general, $H^2(E_{p^m}, \mathbb{F}_p)$ is a vector space over $\mathbb{F}_p$ of dimension $\frac{1}{2}m(m+1)$. When $p = 2$ and $m = 2$ this is the result $H^2(Z_2 \times Z_2, \mathbb{Z}/2\mathbb{Z}) \cong (\mathbb{Z}/2\mathbb{Z})^3$ above.

Crossed Product Algebras and the Brauer Group

Suppose F is a field. Recall that an F -algebra B is a ring containing the field F in its center and the identity of B is the identity of F , cf. Section 10.1.

Definition. An F -algebra A is said to be *simple* if A contains no nontrivial proper (two sided) ideals. A *central simple F -algebra* A is a simple F -algebra whose center is F .

Among the easiest central simple F -algebras are the matrix algebras $M_n(F)$ of $n \times n$ matrices with coefficients in F .

If K/F is a finite Galois extension of fields with Galois group $G = \text{Gal}(K/F)$, then we can use the normalized 2-cocycles in $Z^2(G, K^\times)$ to construct certain central simple K -algebras. The construction of these algebras from 2-cocycles and their classification in terms of $H^2(G, K^\times)$ (cf. Theorem 42 below) are important applications of cohomological methods in number theory. Their construction in the case when G is cyclic was one of the precursors leading to the development of abstract cohomology.

Suppose $f = \{a_{\sigma, \tau}\}_{\sigma, \tau \in G}$ is a normalized 2-cocycle in $Z^2(G, K^\times)$. Let B_f be the vector space over L having basis u_σ for $\sigma \in G$:

$$B_f = \left\{ \sum_{\sigma \in G} \alpha_\sigma u_\sigma \mid \alpha_\sigma \in K \right\}. \quad (17.39)$$

Define a multiplication on B_f by

$$u_\sigma \alpha = \sigma(\alpha) u_\sigma \quad u_\sigma u_\tau = a_{\sigma, \tau} u_{\sigma\tau} \quad (17.40)$$

for $\alpha \in L$ and $\sigma, \tau \in G$. The second equation shows that the $a_{\sigma, \tau}$ give a “factor set” for the elements u_σ in B_f and is one reason this terminology is used. Using this multiplication we find

$$(u_\sigma u_\tau) u_\rho = a_{\sigma, \tau} a_{\sigma\tau, \rho} u_{\sigma\tau\rho} \quad \text{and} \quad u_\sigma (u_\tau u_\rho) = \sigma(a_{\tau, \rho}) a_{\sigma, \tau\rho} u_{\sigma\tau\rho}.$$

Since $a_{\sigma, \tau} a_{\sigma\tau, \rho} = \sigma(a_{\tau, \rho}) a_{\sigma, \tau\rho}$ is the multiplicative form of the cocycle condition (26), it follows that the multiplication defined in (40) is associative.

Since the cocycle is normalized we have $a_{1, \sigma} = a_{\sigma, 1} = 1$ for all $\sigma \in G$ and it follows from (40) that the element u_1 is an identity in B_f . Identifying K with the elements αu_1 in B_f , we see that B_f is an F -algebra containing the field K and having dimension n^2 over F if $n = [K : F] = |G|$.

Proposition 40. The F -algebra B_f with K -vector space basis u_σ in (39) and multiplication defined by (40) is a central simple F -algebra.

Proof: It remains to show that the center of B_f is F and that B_f contains no nonzero proper ideals. Suppose $x = \sum_{\sigma \in G} \alpha_\sigma u_\sigma$ is an element in the center of B_f . Then $x\beta = \beta x$ for $\beta \in K$ shows that $\sigma(\beta) = \beta$ if $\alpha_\sigma \neq 0$. Since there is an element $\beta \in K$ not fixed by σ for any $\sigma \neq 1$, this shows that $\alpha_\sigma = 0$ for all $\sigma \neq 1$, so $x = \alpha_1 u_1$. Then $xu_\tau = u_\tau x$ if and only if $\tau(\alpha_1) = \alpha_1$, so if this is true for all τ then we must have $\alpha_1 = a \in K$. Hence $x = au_1$ and the center of B_f is F .

To show that B_f is simple, suppose I is a nonzero ideal in B_f and let

$$x = \alpha_{\sigma_1} u_{\sigma_1} + \cdots + \alpha_{\sigma_m} u_{\sigma_m}$$

be a nonzero element of I with the minimal number m of nonzero terms. If $m > 1$ there is an element $\beta \in K^\times$ with $\sigma_m(\beta) \neq \sigma_{m-1}(\beta)$. Then the element $x - \sigma_m(\beta) x \beta^{-1}$ would be an element of the ideal I with the nonzero element $(1 - \sigma_m(\beta) \sigma_{m-1}(\beta)^{-1}) \alpha_{\sigma_{m-1}}$ as coefficient of $u_{\sigma_{m-1}}$, and would have fewer nonzero terms than x since the coefficient of u_{σ_m} is 0. It follows that $m = 1$ and $x = \alpha u_\sigma$ for some $\alpha \in K$ and some σ . This element is a unit, with inverse $\sigma^{-1}(\alpha^{-1}) u_{\sigma^{-1}}$, so $I = B_f$, completing the proof.

Definition. The central simple F -algebra B_f defined by (39) and (40) is called the *crossed product algebra* for the factor set $\{a_{\sigma,\tau}\}$.

If $f' = a'_{\sigma,\tau}$ is a normalized cocycle in the same cohomology class in $H^2(G, K^\times)$ as $a_{\sigma,\tau}$ then there are elements $b_\sigma \in K^\times$ with

$$a'_{\sigma,\tau} = a_{\sigma,\tau}(\sigma(b_\tau) b_{\sigma\tau}^{-1} b_\sigma)$$

(the multiplicative form of the coboundary condition (27)). If $B_{f'}$ is the F -algebra with K -basis u_σ defined from this cocycle as in (39) and (40), then the K -vector space homomorphism φ defined by mapping u'_σ to $b_\sigma u_\sigma$ satisfies

$$\begin{aligned} \varphi(u'_\sigma u'_\tau) &= \varphi(a'_{\sigma,\tau} u'_{\sigma\tau}) = a'_{\sigma,\tau} b_{\sigma\tau} u_{\sigma\tau} = b_\sigma \sigma(b_\tau) u_\sigma u_\tau \\ &= (b_\sigma u_\sigma)(b_\tau u_\tau) = \varphi(u'_\sigma) \varphi(u'_\tau). \end{aligned}$$

It follows that φ is an F -algebra isomorphism from $B_{f'}$ to B_f .

We have shown that every cohomology class c in $H^2(G, K^\times)$ defines an isomorphism class of central simple F -algebras, namely the isomorphism class of any crossed product algebra for a normalized cocycle $\{a_{\sigma,\tau}\}$ representing the class c . The next result shows that the trivial cohomology class corresponds to the isomorphism class containing $M_n(F)$.

Proposition 41. The crossed product algebra for the trivial cohomology class in $H^2(G, K^\times)$ is isomorphic to the matrix algebra $M_n(F)$ where $n = [K : F]$.

Proof: If $\alpha \in K$ then multiplication by α defines a linear transformation T_α of K viewed as an n -dimensional vector space over F . Similarly, every automorphism $\sigma \in G$ defines an F -linear transformation T_σ of K , and we may view both T_α and T_σ as

elements of $M_n(F)$ by choosing a basis for K over F . If B_0 denotes the crossed product algebra for the trivial factor set ($a_{\sigma,\tau} = 1$ for all $\sigma, \tau \in G$), consider the additive map $\varphi : B_0 \rightarrow M_n(F)$ defined by $\varphi(\alpha u_\sigma) = T_\alpha T_\sigma$. Since $T_{a\alpha} = aT_\alpha$ for $a \in F$, the map φ is an F -vector space homomorphism. If $x \in K$, we have

$$T_\sigma T_\alpha(x) = T_\sigma(\alpha x) = \sigma(\alpha x) = \sigma(\alpha) \sigma(x) = T_{\sigma(\alpha)} T_\sigma,$$

so $T_\sigma T_\alpha = T_{\sigma(\alpha)} T_\sigma$ as linear transformations on K . It then follows from $u_\sigma u_\tau = u_{\sigma\tau}$ that

$$\begin{aligned}\varphi((\alpha u_\sigma)(\beta u_\tau)) &= \varphi(\alpha\sigma(\beta) u_{\sigma\tau}) = T_{\alpha\sigma(\beta)} T_{\sigma\tau} = T_\alpha T_{\sigma(\beta)} T_\sigma T_\tau \\ &= T_\alpha T_\sigma T_\beta T_\tau = \varphi(\alpha u_\sigma) \varphi(\beta u_\tau)\end{aligned}$$

which shows that φ is an F -algebra homomorphism from B_0 to $M_n(F)$. Since $\ker \varphi$ is an ideal in B_0 and $\varphi \neq 0$, it follows from Proposition 40 that $\ker \varphi = 0$ and φ is an injection. Since both B_0 and $M_n(F)$ have dimension n^2 as vector spaces over F , it follows that φ is an F -algebra isomorphism, proving the proposition.

Example

If $K = \mathbb{C}$ and $F = \mathbb{R}$, then $G = \text{Gal}(\mathbb{C}/\mathbb{R})$ is of order 2 and generated by complex conjugation τ . We have $|H^2(G, \mathbb{C}^\times)| = 2$. The central simple $\mathbb{R}$ -algebra B_0 corresponding to the trivial class is $\mathbb{C}u_1 \oplus \mathbb{C}u_\tau$ with $u_\tau(a + bi) = (a - bi)u_\tau$ and $u_\tau^2 = u_1$. This is isomorphic to the matrix algebra $M_2(\mathbb{R})$ under the map

$$\varphi((a + bi)u_1 + (c + di)u_\tau) = aI + bT_i + cT_\tau + dT_i T_\tau = \begin{pmatrix} a + c & -b + d \\ b + d & a - c \end{pmatrix}.$$

A normalized cocycle f representing the nontrivial cohomology class is defined by the values $a_{1,1} = a_{1,\tau} = a_{\tau,1} = 1$ and $a_{\tau,\tau} = -1$. The corresponding central simple $\mathbb{R}$ -algebra B_f is given by $\mathbb{C}v_1 \oplus \mathbb{C}v_\tau$. The element v_1 is the identity of B_f , and we have the relations $v_\tau(a + bi) = (a - bi)v_\tau$ and $v_\tau^2 = -v_1$. Letting $v_1 = 1$ and $v_\tau = j$ we see that B_f is isomorphic as an $\mathbb{R}$ -algebra to the real Hamilton Quaternions $\mathbb{R} + \mathbb{R}i + \mathbb{R}j + \mathbb{R}k$.

There is a rich theory of simple algebras and we mention without proof the following results. Let A be a central simple F -algebra of finite dimension over F .

- I. If $F \subseteq B \subseteq A$ where B is a simple F -algebra define the *centralizer* B^c of B in A to be the elements of A that commute with all the elements of B . Define the *opposite algebra* B^{opp} to be the set B with opposite multiplication, i.e., the product $b_1 b_2$ in B^{opp} is given by the product $b_2 b_1$ in B . Both B^c and B^{opp} are simple F -algebras and we have
 - a. $(\dim_F B)(\dim_F B^c) = \dim_F A$
 - b. $A \otimes_F B^{opp} \cong M_r(B^c)$ as F -algebras, where $r = \dim_F B$
 - c. $B \otimes_F B^c \cong A$ if B is a central simple F -algebra.
- II. If A' is an Artinian (satisfies D.C.C. on left ideals) simple F -algebra, then $A \otimes_F A'$ is an Artinian simple F -algebra with center $(A')^c$.
- III. We have $A \cong M_r(\Delta)$ for some division ring Δ whose center is F and some integer $r \geq 1$. The division ring Δ and r are uniquely determined by A . The same statement holds for any Artinian simple F -algebra.

The last result is part of Wedderburn's Theorem described in greater detail in the following chapter.

Definition. If A is a central simple F -algebra then a field L containing F is said to *split* A if $A \otimes_F L \cong M_m(L)$ for some $m \geq 1$.

It follows from (II) that every maximal commutative subalgebra of Δ is a field E with $E = E^c = E^{opp}$; if $[E : F] = m$ we obtain $\dim_F \Delta = m^2$. Applying (II) to $A = \Delta$ and $B = E$ we also see that $\Delta \otimes_F E \cong M_m(E)$. It can also be shown that a maximal subfield E of the central simple F -algebra A also satisfies $E = E^c = E^{opp}$ and so again by (II) it follows that $A \otimes_F E \cong M_r(E)$ ($r^2 = \dim_F A$).

If $A = M_r(\Delta)$ then the field L splits A if and only if L splits Δ , as follows. If $\Delta \otimes_F L \cong M_n(L)$ then

$$A \otimes_F L \cong M_r(\Delta) \otimes_F L \cong M_r(\Delta \otimes_F L) \cong M_r(M_n(L)) \cong M_{rn}(L).$$

Conversely if $A \otimes_F L \cong M_n(L)$ then

$$M_n(L) \cong M_r(\Delta) \otimes_F L \cong M_r(\Delta \otimes_F L).$$

By (II) and (III), $\Delta \otimes_F L \cong M_s(\Delta')$ for some division ring Δ' . Together with the previous isomorphism, the uniqueness statement in (III) shows that $\Delta' \cong L$ and then the isomorphism $\Delta \otimes_F L \cong M_s(L)$ shows that L splits Δ .

We see from the discussion above that a maximal commutative subfield of Δ splits both Δ and $A \cong M_r(\Delta)$ for any $r \geq 1$. It is not too difficult to show from this that every central simple F -algebra of finite dimension over F can be split by a finite Galois extension of F .

Applying (I) by taking A to be the crossed product algebra B_f and taking $B = K$ shows that $K = K^c = K^{opp}$ and $B_f \otimes_F K \cong M_n(K)$. In particular, the crossed product algebras B_f are always split by K .

Example

In the example of the Hamilton Quaternions above we have $B_f \otimes_{\mathbb{R}} \mathbb{C} \cong M_2(\mathbb{C})$. We have $B_f \otimes_{\mathbb{R}} \mathbb{C} = \mathbb{C} + \mathbb{C}i + \mathbb{C}j + \mathbb{C}k$ and an explicit isomorphism φ to $M_2(\mathbb{C})$ is given by

$$\varphi(i) = \begin{pmatrix} \sqrt{-1} & 0 \\ 0 & -\sqrt{-1} \end{pmatrix} \quad \varphi(j) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

and extending $\mathbb{C}$ -linearly.

By (III) every central simple F -algebra A is isomorphic as an F -algebra to $M_r(\Delta)$ for some division ring Δ uniquely determined up to F -isomorphism, called the *division ring part* of A .

Definition. Two central simple F -algebras A and B are *similar* if $A \cong M_r(\Delta)$ and $B \cong M_s(\Delta)$ for the same division ring Δ , i.e., if A and B have the same division ring parts.

Let $[A]$ denote the similarity class of A . By (II), if A and B are central simple F -algebras then $A \otimes_F B$ is again a central simple F -algebra, so we may define a multiplication on similarity classes by $[A][B] = [A \otimes_F B]$. The class $[F]$ is an identity for this multiplication and associativity of the tensor product shows that the multiplication is associative. By (Ib) applied with $B = A$ (so then $B^c = F$ since A is central) we have $[A][A^{opp}] = [F]$, so inverses exist with this multiplication.

Definition. The group of similarity classes of central simple F -algebras with multiplication $[A][B] = [A \otimes_F B]$ is called the *Brauer group* of F and is denoted $Br(F)$.

If L is any extension field of F then by (II) the algebra $A \otimes_F L$ is a central simple L -algebra. It is easy to check that the map $[A] \rightarrow [A \otimes_F L]$ is a well defined homomorphism from $Br(F)$ to $Br(L)$. The kernel of this homomorphism consists of the classes of the algebras A with $A \otimes_F L \cong M_m(L)$ for some $m \geq 1$, i.e., the algebras A that are split by L .

Definition. If L/F is a field extension then the *relative Brauer group* $Br(L/F)$ is the group of similarity classes of central simple F -algebras that are split by L . Equivalently, $Br(L/F)$ is the kernel of the homomorphism $[A] \rightarrow [A \otimes_F L]$ from $Br(F)$ to $Br(L)$.

The following theorem summarizes some major results in this area and shows the fundamental connection between Brauer groups and the crossed product algebras constructed above.

Theorem 42. Suppose K/F is a Galois extension of degree n with $G = \text{Gal}(K/F)$.

- (1) The central simple F -algebra A with $\dim_F A = n^2$ is split by K if and only if $A \otimes_F K \cong M_n(K)$ if and only if A is isomorphic to a crossed product algebra B_f as in (39) and (40).
- (2) There is a bijection between the F -isomorphism classes of central simple F -algebras A with $A \otimes_F K \cong M_n(K)$ and the elements of $H^2(G, K^\times)$. Under this bijection the class $c \in H^2(G, K^\times)$ containing the normalized cocycle f corresponds to the isomorphism class of the crossed product algebra B_f defined in (39) and (40), and the trivial cohomology class corresponds to $M_n(F)$.
- (3) Every central simple F -algebra of finite dimension over F and split by K is similar to one of dimension n^2 split by K . The bijection in (2) also establishes a bijection between $Br(K/F)$ and $H^2(G, K^\times)$ which is also an isomorphism of groups.
- (4) There is a bijection between the collection of F -isomorphism classes of central simple division algebras over F that are split by K and $H^2(G, K^\times)$.

As previously mentioned, every central simple F -algebra of finite dimension over F can be split by some finite Galois extension of F , and it follows that

$$Br(F) = \bigcup_K Br(K/F)$$

where the union is over all finite Galois extensions of F . It follows that there is a bijection between $Br(F)$ and $H^2(\text{Gal}(F^s/F), (F^s)^\times)$ where F^s denotes a separable algebraic closure of F . Here $\text{Gal}(F^s/F)$ is considered as a profinite group and the cohomology group refers to continuous Galois cohomology.

One consequence of this result and Theorem 42 is that a full set of representatives for the F -isomorphism classes of central simple division algebras Δ over F can be obtained from the division algebra parts of the crossed product algebras for finite Galois extensions of F . Those division algebras that are split over K occur for the crossed product algebras for K/F .

Example

Since $H^2(\text{Gal}(\mathbb{F}_{q^d}/\mathbb{F}_q), \mathbb{F}_q^\times) = 0$ (cf. Exercise 10), we have $Br(\mathbb{F}_{q^d}/\mathbb{F}_q) = 0$ and hence also $Br(\mathbb{F}_q) = 0$. As a consequence, every finite division algebra is a field (cf. Exercise 13 in Section 13.6 for a direct proof), and every finite central simple algebra $\mathbb{F}_q$ -algebra is isomorphic to a full matrix ring $M_r(\mathbb{F}_q)$.

EXERCISES

- Let $A = \{1, a, b, c\}$ be the Klein 4-group and let $G = \langle g \rangle$ be the cyclic group of order 2 acting trivially on A .
 - Prove that $|C^2(G, A)| = 2^8$.
 - Show that coboundaries are constant functions, and deduce that $|B^2(G, A)| = 4$.
 - Use the cocycle condition to show that $|Z^2(G, A)| \leq 2^4$.
 - If $E = Z_4 \times Z_2 = \langle x \rangle \times \langle y \rangle$, prove that the extensions $1 \rightarrow A \xrightarrow{i} E \xrightarrow{\pi} G \rightarrow 1$ defined by $\pi(x) = g, \pi(y) = 1$ and $\iota_1(a) = x^2, \iota_1(b) = y$ (respectively, $\iota_2(b) = x^2, \iota_2(a) = y$, and $\iota_3(c) = x^2, \iota_3(a) = y$), together with the split extension $Z_2 \times Z_2 \times Z_2$ give 4 inequivalent extensions of Z_2 by the Klein 4-group. Deduce that $H^2(G, A)$ has order 4 by explicitly exhibiting the corresponding cocycles.
- Let $A = \mathbb{Z}/4\mathbb{Z}$ and let G be the cyclic group of order 2 acting trivially on A .
 - Prove that $|C^2(G, A)| = 2^8$.
 - Use the coboundary condition to show that $|B^2(G, A)| = 2^3$.
 - Use the cocycle condition to show that $|Z^2(G, A)| \leq 2^4$.
 - Show that $|H^2(G, A)| = 2$ by exhibiting two inequivalent extensions of G by A and their corresponding cocycles.
- Let $A = \mathbb{Z}/4\mathbb{Z}$ and let G be the cyclic group of order 2 acting by inversion on A .
 - Show that there are four coboundaries and that only the zero coboundary is normalized.
 - Prove by a direct computation of cocycle and coboundary groups that $|H^2(G, A)| = 2$.
 - Exhibit two distinct cohomology classes and their corresponding extension groups.
 - Show that for a given extension of G by A with extension group isomorphic to D_8 there are four normalized sections, all of which have the zero 2-cocycle as their factor set.
 - Show that for a given extension of G by A with extension group isomorphic to Q_8 there are sixteen sections, four of which are normalized, and all of the latter have the same factor set.
- For a non-normalized 2-cocycle f one defines the extension group E_f on the set $A \times G$ by the same binary operation in equation (34). Verify two of the group axioms in this case by showing that identity is now $(-f(1, 1), 1)$ and inverses are given by

$$(a, x)^{-1} = (-x^{-1} \cdot a - f(x^{-1}, x) - f(1, 1), x^{-1}).$$

(Verification of the associative law is essentially the same as for normalized 2-cocycles.) Prove also that the set $A^{**} = \{(a - f(1, 1), 1) \mid a \in A\}$ is a subgroup of E_f and the map $\iota^{**}: a \mapsto (a - f(1, 1), 1)$ is an isomorphism from A to A^{**} . Show that this extension E_f , with the injection ι^{**} and the usual projection map π^* onto G , is equivalent to an extension derived from a normalized cocycle in the same class as f .

- Show that the set of equivalences of a given extension $1 \rightarrow A \xrightarrow{\iota} E \xrightarrow{\pi} G \rightarrow 1$ with itself form a group under composition, and that this group is isomorphic to the stability group

$\text{Stab}(1 \trianglelefteq \iota(A) \trianglelefteq E)$. (Thus Proposition 31 implies $Z^1(G, A)$ is the group of equivalences of the extension with itself).

6. (*Gaschütz's Theorem*) Let p be a prime, let A be an abelian normal p -subgroup of a finite group G , and let P be a Sylow p -subgroup of G . Prove that G is a split extension of G/A by A if and only if P is a split extension of P/A by A . (Note that $A \leq P$ by Exercise 37 in Section 4.5). [Use Sylow's Theorem to show if G splits over A then so too does P . Conversely, show that a normalized 2-cocycle associated to the extension of P/A by A via Theorem 36 is the image of a normalized 2-cocycle in $H^2(G/A, A)$ under the restriction homomorphism $\text{Res} : H^2(G/A, A) \rightarrow H^2(P/A, A)$. Then use Proposition 26 and the fact that multiplication by $|G : P|$ is an automorphism of A .]
7. (a) Prove that $H^2(A_4, \mathbb{Z}/2\mathbb{Z}) \neq 0$ by exhibiting a nonsplit extension of A_4 by a cyclic group of order 2. [See Exercise 11, Section 4.5.]
 (b) Prove that $H^2(A_5, \mathbb{Z}/2\mathbb{Z}) \neq 0$ by showing that $SL_2(\mathbb{F}_5)$ is a nonsplit extension of A_5 by a cyclic group of order 2. [Use Propositions 21 and 23 in Section 4.5.]
8. The *Schur multiplier* of a finite group G is defined as the group $H^2(G, \mathbb{C}^\times)$, where the multiplicative group $\mathbb{C}^\times$ of complex numbers is a trivial G -module. Prove that the Schur multiplier is a finite group. [Show that every cohomology class contains a cocycle whose values lie in the n^{th} roots of unity, where $n = |G|$, as follows: If f is any cocycle then by Corollary 27, $f^n \in B^2(G, \mathbb{C}^\times)$. Define $k \in C^2(G, \mathbb{C}^\times)$ by $k(g_1, g_2) = f(g_1, g_2)^{1/n}$ (take any n^{th} roots). Show that $k \in B^2(G, \mathbb{C}^\times)$ and fk^{-1} takes values in the group of n^{th} roots of 1.]
9. Use the classification of the extensions of the Klein 4-group by Z_2 in the example following Theorem 39 to prove the following (in the notation of that example):
 (a) There is an (outer) automorphism of $Z_4 \times Z_2$ which interchanges the cosets Ax and Axy and fixes the coset Ay .
 (b) There is an outer automorphism of D_8 which interchanges the cosets As and Asr and fixes the coset Ar .
10. Suppose $\mathbb{F}_q$ is a finite field with $G = \text{Gal}(\mathbb{F}_{q^d}/\mathbb{F}_q) = \langle \sigma_q \rangle$ where σ_q is the Frobenius automorphism, and let N be the usual norm element for the cyclic group G .
 (a) Use Hilbert's Theorem 90 to prove that $|_N(\mathbb{F}_{q^d}^\times)| = (q^d - 1)/(q - 1)$, and deduce that the norm map from $\mathbb{F}_{q^d}$ to $\mathbb{F}_q$ is surjective.
 (b) Prove that $H^n(G, \mathbb{F}_{q^d}^\times) = 0$ for all $n \geq 1$.

Part VI

INTRODUCTION TO THE REPRESENTATION THEORY OF FINITE GROUPS

The final two chapters are an introduction to the representation theory of finite groups together with some applications. We have already seen in Part I how actions of groups on sets, namely permutation representations, are a fundamental tool for unravelling the structure of groups. Cayley's Theorem and Sylow's Theorem as well as many of the results and applications in Sections 6.1 and 6.2 are based on groups acting on sets. The chapter on Galois Theory developed one of the most beautiful correspondences in mathematics where the action of a group as automorphisms of a field gives rise to a correspondence between the lattice of subgroups of the Galois group and the lattice of subfields of a Galois extension of fields. In these final two chapters we study groups acting as linear transformations on vector spaces. We shall be primarily interested in utilizing these linear actions to provide information about the groups themselves.

In Part III we saw that modules are the “representation objects” for rings in the sense that the axioms for an R -module specify a “ring action” of R on some abelian group M which preserves the abelian group structure of M . In the case where M was an $F[x]$ -module, x acted as a linear transformation from the vector space M to itself. In Chapter 12 the classification of finitely generated modules over Principal Ideal Domains gave us a great deal of information about these linear transformations of M (e.g., canonical forms). In Chapter 16 we used the ideal structure in Dedekind Domains to generalize the results of Chapter 12 to the classification of finitely generated modules over such domains. In this part we follow a process similar to the study of $F[x]$ -modules, replacing the polynomial ring with the group ring FG of G and classifying all finitely generated FG -modules for certain fields F (Wedderburn's Theorem). We then use this classification to derive some results about finite groups such as Burnside's Theorem on the solvability of groups of order $p^a q^b$ in Chapter 19.

Representation Theory and Character Theory

18.1 LINEAR ACTIONS AND MODULES OVER GROUP RINGS

For the remainder of the book the groups we consider will be finite groups, unless explicitly mentioned otherwise. Throughout this section F is a field and G is a finite group. We first introduce the basic terminology. Recall that if V is a vector space over F , then $GL(V)$ is the group of nonsingular linear transformations from V to itself (under composition), and if $n \in \mathbb{Z}^+$, then $GL_n(F)$ is the group of invertible $n \times n$ matrices with entries from F (under matrix multiplication).

Definition. Let G be a finite group, let F be a field and let V be a vector space over F .

- (1) A *linear representation* of G is any homomorphism from G into $GL(V)$. The *degree* of the representation is the dimension of V .
- (2) Let $n \in \mathbb{Z}^+$. A *matrix representation* of G is any homomorphism from G into $GL_n(F)$.
- (3) A linear or matrix representation is *faithful* if it is injective.
- (4) The *group ring* of G over F is the set of all formal sums of the form

$$\sum_{g \in G} \alpha_g g, \quad \alpha_g \in F$$

with componentwise addition and multiplication $(\alpha g)(\beta h) = (\alpha\beta)(gh)$ (where α and β are multiplied in F and gh is the product in G) extended to sums via the distributive law (cf. Section 7.2).

Unless we are specifically discussing permutation representations the term “representation” will always mean “linear representation.” When we wish to emphasize the field F we shall say F -representation, or representation of G on V over F .

Recall that if V is a finite dimensional vector space of dimension n , then by fixing a basis of V we obtain an isomorphism $GL(V) \cong GL_n(F)$. In this way any linear representation of G on a finite dimensional vector space gives a matrix representation and vice versa. For the most part our linear representations will be of finite degree and we shall pass freely between linear representations and matrix representations (specifying a

basis when we wish to give an explicit correspondence between the two). Furthermore, given a linear representation $\varphi : G \rightarrow GL(V)$ of finite degree, a corresponding matrix representation provides numerical invariants (such as the determinant of $\varphi(g)$ for $g \in G$) which are independent of the choice of basis giving the isomorphism between $GL(V)$ and $GL_n(F)$. The exploitation of such invariants will be fundamental to our development.

Before giving examples of representations we recall the group ring FG in greater detail (group rings were introduced in Section 7.2, and some notation and examples were discussed in that section). Suppose the elements of G are $g_1, g_2, \dots, g_n$. Each element of FG is of the form

$$\sum_{i=1}^n \alpha_i g_i, \quad \alpha_i \in F.$$

Two formal sums¹ are equal if and only if all corresponding coefficients of group elements are equal. Addition and multiplication in FG are defined as follows:

$$\begin{aligned} \sum_{i=1}^n \alpha_i g_i + \sum_{i=1}^n \beta_i g_i &= \sum_{i=1}^n (\alpha_i + \beta_i) g_i \\ \left(\sum_{i=1}^n \alpha_i g_i \right) \left(\sum_{i=1}^n \beta_i g_i \right) &= \sum_{k=1}^n \left(\sum_{\substack{i,j \\ g_i g_j = g_k}} \alpha_i \beta_j \right) g_k \end{aligned}$$

where addition and multiplication of the coefficients α_i and β_j is performed in F . Note that by definition of multiplication,

FG is a commutative ring if and only if G is an abelian group.

The group G appears in FG (identifying g_i with $1g_i$) and the field F appears in FG (identifying β with βg_1 , where g_1 is the identity of G). Under these identifications

$$\beta \left(\sum_{i=1}^n \alpha_i g_i \right) = \sum_{i=1}^n (\beta \alpha_i) g_i, \quad \text{for all } \beta \in F.$$

In this way

FG is a vector space over F with the elements of G as a basis.

In particular, FG is a vector space over F of dimension equal to $|G|$. The elements of F commute with all elements of FG , i.e., F is in the *center* of FG . When we wish to emphasize the latter two properties we shall say that FG is an *F -algebra* (in general, an *F -algebra* is a ring R which contains F in its center, so R is both a ring and an F -vector space).

Note that the operations in FG are similar to those in the F -algebra $F[x]$ (although $F[x]$ is infinite dimensional over F). In some works FG is denoted by $F[G]$, although the latter notation is currently less prevalent.

¹The formal sum displayed above is a way of writing the function from G to F which takes the value α_i on the group element g_i . This same “formality” was used in the construction of free modules (see Theorem 6 in Section 10.3).

Examples

- (1) If $G = \langle g \rangle$ is cyclic of order $n \in \mathbb{Z}^+$, then the elements of FG are of the form

$$\sum_{i=0}^{n-1} \alpha_i g^i.$$

The map $F[x] \rightarrow F\langle g \rangle$ which sends x^k to g^k for all $k \geq 0$ extends by F -linearity to a surjective ring homomorphism with kernel equal to the ideal generated by $x^n - 1$. Thus

$$F\langle g \rangle \cong F[x]/(x^n - 1).$$

This is an isomorphism of F -algebras, i.e., is a ring isomorphism which is F -linear.

- (2) Under the notation of the preceding example let $r = 1 + g + g^2 + \cdots + g^{n-1}$, so r is a nonzero element of $F\langle g \rangle$. Note that $rg = g + g^2 + \cdots + g^{n-1} + 1 = r$, hence $r(1 - g) = 0$. Thus the ring $F\langle g \rangle$ contains zero divisors (provided $n > 1$). More generally, if G is any group of order > 1 , then for any nonidentity element $g \in G$, $F\langle g \rangle$ is a subring of FG , so FG also contains zero divisors.
- (3) Let $G = S_3$ and $F = \mathbb{Q}$. The elements $r = 5(1\ 2) - 7(1\ 2\ 3)$ and $s = -4(1\ 2\ 3) + 12(1\ 3\ 2)$ are typical members of $\mathbb{Q}S_3$. Their sum and product are seen to be

$$r + s = 5(1\ 2) - 11(1\ 2\ 3) + 12(1\ 3\ 2)$$

$$rs = -20(2\ 3) + 28(1\ 3\ 2) + 60(1\ 3) - 84$$

(recall that products (compositions) of permutations are computed from right to left). An explicit example of a sum and product of two elements in the group ring $\mathbb{Q}D_8$ appears in Section 7.2.

Before giving specific examples of representations we discuss the correspondence between representations of G and FG -modules (after which we can simultaneously give examples of both). This discussion closely parallels the treatment of $F[x]$ -modules in Section 10.1.

Suppose first that $\varphi : G \rightarrow GL(V)$ is a representation of G on the vector space V over F . As above, write $G = \{g_1, \dots, g_n\}$, so for each $i \in \{1, \dots, n\}$, $\varphi(g_i)$ is a linear transformation from V to itself. Make V into an FG -module by defining the action of a ring element on an element of V as follows:

$$\left(\sum_{i=1}^n \alpha_i g_i \right) \cdot v = \sum_{i=1}^n \alpha_i \varphi(g_i)(v), \quad \text{for all } \sum_{i=1}^n \alpha_i g_i \in FG, \ v \in V.$$

We verify a special case of axiom 2(b) of a module (see Section 10.1) which shows precisely where the fact that φ is a group homomorphism is needed:

$$\begin{aligned} (g_i g_j) \cdot v &= \varphi(g_i g_j)(v) && \text{(by definition of the action)} \\ &= (\varphi(g_i) \circ \varphi(g_j))(v) && \text{(since } \varphi \text{ is a group homomorphism)} \\ &= \varphi(g_i)(\varphi(g_j)(v)) && \text{(by definition of a composition of linear transformations)} \\ &= g_i \cdot (g_j \cdot v) && \text{(by definition of the action).} \end{aligned}$$

This argument extends by linearity to arbitrary elements of FG to prove that axiom 2(b) of a module holds in general. It is an exercise to check that the remaining module axioms hold.

Note that F is a subring of FG and the action of the field element α on a vector is the same as the action of the ring element $\alpha 1$ on a vector i.e., the FG -module action extends the F action on V .

Suppose now that conversely we are given an FG -module V . We obtain an associated vector space over F and representation of G as follows. Since V is an FG -module, it is an F -module, i.e., it is a vector space over F . Also, for each $g \in G$ we obtain a map from V to V , denoted by $\varphi(g)$, defined by

$$\varphi(g)(v) = g \cdot v \quad \text{for all } v \in V,$$

where $g \cdot v$ is the given action of the ring element g on the element v of V . Since the elements of F commute with each $g \in G$ it follows by the axioms for a module that for all $v, w \in V$ and all $\alpha, \beta \in F$ we have

$$\begin{aligned} \varphi(g)(\alpha v + \beta w) &= g \cdot (\alpha v + \beta w) \\ &= g \cdot (\alpha v) + g \cdot (\beta w) \\ &= \alpha(g \cdot v) + \beta(g \cdot w) \\ &= \alpha\varphi(g)(v) + \beta\varphi(g)(w), \end{aligned}$$

that is, for each $g \in G$, $\varphi(g)$ is a linear transformation. Furthermore, it follows by axiom 2(b) of a module that

$$\varphi(g_i g_j)(v) = (\varphi(g_i) \circ \varphi(g_j))(v)$$

(this is essentially the calculation above with the steps reversed). This proves that φ is a group homomorphism (in particular, $\varphi(g^{-1}) = \varphi(g)^{-1}$, so every element of G maps to a nonsingular linear transformation, i.e., $\varphi : G \rightarrow GL(V)$).

This discussion shows there is a bijection between FG -modules and pairs (V, φ) :

$$\left\{ \begin{array}{l} V \text{ an } FG\text{-module} \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} V \text{ a vector space over } F \\ \text{and} \\ \varphi : G \rightarrow GL(V) \text{ a representation} \end{array} \right\}.$$

Giving a representation $\varphi : G \rightarrow GL(V)$ on a vector space V over F is therefore equivalent to giving an FG -module V . Under this correspondence we shall say that the module V *affords* the representation φ of G .

Recall from Section 10.1 that if a vector space M is made into an $F[x]$ -module via the linear transformation T , then the $F[x]$ -submodules of M are precisely the T -stable subspaces of M . In the current situation if V is an FG -module affording the representation φ , then a subspace U of V is called *G-invariant* or *G-stable* if $g \cdot u \in U$ for all $g \in G$ and all $u \in U$ (i.e., if $\varphi(g)(u) \in U$ for all $g \in G$ and all $u \in U$). It follows easily that

the FG -submodules of V are precisely the G -stable subspaces of V .

Examples

- (1) Let V be a 1-dimensional vector space over F and make V into an FG -module by letting $gv = v$ for all $g \in G$ and $v \in V$. This module affords the representation $\varphi : G \rightarrow GL(V)$ defined by $\varphi(g) = I =$ the identity linear transformation, for all $g \in G$. The corresponding matrix representation (with respect to any basis of V) is the homomorphism of G into $GL_1(F)$ which sends every group element to the 1×1 identity matrix. We shall henceforth refer to this as the *trivial representation* of G . The trivial representation has degree 1 and if $|G| > 1$, it is not faithful.
- (2) Let $V = FG$ and consider this ring as a left module over itself. Then V affords a representation of G of degree equal to $|G|$. If we take the elements of G as a basis of V , then each $g \in G$ permutes these basis elements under the left regular permutation representation:

$$g \cdot gi = ggi.$$

With respect to this basis of V the matrix of the group element g has a 1 in row i and column j if $ggi = gi$, and has 0's in all other positions. This (linear or matrix) representation is called the *regular representation* of G . Note that each nonidentity element of G induces a nonidentity permutation on the basis of V so the regular representation is always faithful.

- (3) Let $n \in \mathbb{Z}^+$, let $G = S_n$ and let V be an n -dimensional vector space over F with basis $e_1, e_2, \dots, e_n$. Let S_n act on V by defining for each $\sigma \in S_n$

$$\sigma \cdot e_i = e_{\sigma(i)}, \quad 1 \leq i \leq n$$

i.e., σ acts by permuting the subscripts of the basis elements. This provides an (injective) homomorphism of S_n into $GL(V)$ (i.e., a faithful representation of S_n of degree n), hence makes V into an FS_n -module. As in the preceding example, the matrix of σ with respect to the basis $e_1, \dots, e_n$ has a 1 in row i and column j if $\sigma \cdot e_j = e_i$ (and has 0 in all other entries). Thus σ has a 1 in row i and column j if $\sigma(j) = i$.

For an example of the ring action, consider the action of FS_3 on the 3-dimensional vector space over F with basis e_1, e_2, e_3 . Let σ be the transposition (1 2), let τ be the 3-cycle (1 2 3) and let $r = 2\sigma - 3\tau \in FS_3$. Then

$$\begin{aligned} r \cdot (\alpha e_1 + \beta e_2 + \gamma e_3) &= 2(\alpha e_{\sigma(1)} + \beta e_{\sigma(2)} + \gamma e_{\sigma(3)}) - 3(\alpha e_{\tau(1)} + \beta e_{\tau(2)} + \gamma e_{\tau(3)}) \\ &= 2(\alpha e_2 + \beta e_1 + \gamma e_3) - 3(\alpha e_2 + \beta e_3 + \gamma e_1) \\ &= (2\beta - 3\gamma)e_1 - \alpha e_2 + (2\gamma - 3\beta)e_3. \end{aligned}$$

- (4) If $\psi : H \rightarrow GL(V)$ is any representation of H and $\varphi : G \rightarrow H$ is any group homomorphism, then the composition $\psi \circ \varphi$ is a representation of G . For example, let V be the FS_n -module of dimension n described in the preceding example. If $\pi : G \rightarrow S_n$ is any permutation representation of G , the composition of π with the representation above gives a linear representation of G . In other words, V becomes an FG -module under the action

$$g \cdot e_i = e_{\pi(g)(i)}, \quad \text{for all } g \in G.$$

Note that the regular representation, (2), is just the special case of this where $n = |G|$ and π is the left regular permutation representation of G .

- (5) Any homomorphism of G into the multiplicative group $F^\times = GL_1(F)$ is a degree 1 (matrix) representation. For example, suppose $G = \langle g \rangle \cong Z_n$ is the cyclic group of order n and ζ is a fixed n^{th} root of 1 in F . Let $g^i \mapsto \zeta^i$, for all $i \in \mathbb{Z}$. This representation of $\langle g \rangle$ is a faithful representation if and only if ζ is a primitive n^{th} root of 1.

- (6) In many situations it is easier to specify an explicit matrix representation of a group G rather than to exhibit an FG -module. For example, recall that the dihedral group D_{2n} has the presentation

$$D_{2n} = \langle r, s \mid r^n = s^2 = 1, rs = sr^{-1} \rangle.$$

If R and S are any matrices satisfying the relations $R^n = S^2 = I$ and $RS = SR^{-1}$ then the map $r \mapsto R$ and $s \mapsto S$ extends uniquely to a homomorphism from D_{2n} to the matrix group generated by R and S , hence gives a representation of D_{2n} . An explicit example of matrices $R, S \in M_2(\mathbb{R})$ may be obtained as follows. If a regular n -gon is drawn on the x, y plane centered at the origin with the line $y = x$ as one of its lines of symmetry then the matrix R that rotates the plane through $2\pi/n$ radians and the matrix S that reflects the plane about the line $y = x$ both send this n -gon onto itself. It follows that these matrices act as symmetries of the n -gon and so satisfy the above relations. These matrices are readily computed (cf. Exercise 25, Section 1.6) and so the maps

$$r \mapsto R = \begin{pmatrix} \cos 2\pi/n & -\sin 2\pi/n \\ \sin 2\pi/n & \cos 2\pi/n \end{pmatrix} \quad \text{and} \quad s \mapsto S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

extend uniquely to a (degree 2) representation of D_{2n} into $GL_2(\mathbb{R})$. Since the matrices R and S have orders n and 2 respectively, it follows that they generate a subgroup of $GL_2(\mathbb{R})$ of order $2n$ and hence this representation is faithful.

- (7) By using the usual generators and relations for the quaternion group

$$Q_8 = \langle i, j \mid i^4 = j^4 = 1, i^2 = j^2, i^{-1}ji = j^{-1} \rangle$$

one may similarly obtain (cf. Exercise 26, Section 1.6) a representation φ from Q_8 to $GL_2(\mathbb{C})$ defined by

$$\varphi(i) = \begin{pmatrix} \sqrt{-1} & 0 \\ 0 & -\sqrt{-1} \end{pmatrix} \quad \text{and} \quad \varphi(j) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

This representation of Q_8 is faithful.

- (8) A 4-dimensional representation of the quaternion group Q_8 may be obtained from the real Hamilton quaternions, $\mathbb{H}$ (cf. Section 7.1). The group Q_8 is a subgroup of the multiplicative group of units of $\mathbb{H}$ and each of the elements of Q_8 acts by left multiplication on the 4-dimensional real vector space $\mathbb{H}$. Since the real numbers are in the center of $\mathbb{H}$ (i.e., since $\mathbb{H}$ is an $\mathbb{R}$ -algebra), left multiplication is $\mathbb{R}$ -linear. This linear action thus gives a homomorphism from Q_8 into $GL_4(\mathbb{R})$. One can easily write out the explicit matrices of each of the elements of Q_8 with respect to the basis $1, i, j, k$ of $\mathbb{H}$. For example, left multiplication by i acts by $1 \mapsto i, i \mapsto -1, j \mapsto k$ and $k \mapsto -j$ and left multiplication by j acts by $1 \mapsto j, i \mapsto -k, j \mapsto -1$ and $k \mapsto i$ so

$$i \mapsto \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad \text{and} \quad j \mapsto \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}.$$

This representation of Q_8 is also faithful.

- (9) Suppose that H is a normal subgroup of the group G and suppose that H is an elementary abelian p -group for some prime p . Then $V = H$ is a vector space over $\mathbb{F}_p$, where the scalar a acts on the vector v by $av = v^a$ (see Section 10.1). The action of each element of G by conjugation on V is $\mathbb{F}_p$ -linear because $gv^a g^{-1} = (gvg^{-1})^a$ and this action of G on V makes V into an $\mathbb{F}_p G$ -module (the automorphisms of elementary abelian p -groups were discussed in Sections 4.4 and 10.1). The kernel of

this representation is the set of elements of G that commute with every element of H , $C_G(H)$ (which always contains the abelian group H itself). Thus the action of a group on subsets of itself often affords linear representations over finite fields. Representations of groups over finite fields are called *modular representations* and these are fundamental to the study of the internal structure of groups.

- (10) For an example of an FG -submodule, let $G = S_n$ and let V be the FS_n -module described in Example 3. Let N be the subspace of V consisting of vectors all of whose coordinates are equal, i.e.,

$$N = \{\alpha_1 e_1 + \alpha_2 e_2 + \cdots + \alpha_n e_n \mid \alpha_1 = \alpha_2 = \cdots = \alpha_n\}$$

(this is a 1-dimensional S_n -stable subspace). Each $\sigma \in S_n$ fixes each vector in N so the submodule N affords the trivial representation of S_n . As an exercise, one may show that if $n \geq 3$ then N is the *unique* 1-dimensional subspace of V which is S_n -stable, i.e., N is the unique 1-dimensional FS_n -submodule (N is called the *trace* submodule of FS_n).

Another FS_n -submodule of V is the subspace I of all vectors whose coordinates sum to zero:

$$I = \{\alpha_1 e_1 + \alpha_2 e_2 + \cdots + \alpha_n e_n \mid \alpha_1 + \alpha_2 + \cdots + \alpha_n = 0\}.$$

Again I is an S_n -stable subspace (since each $\sigma \in S_n$ permutes the coordinates of each vector in V , each σ leaves the sum of the coefficients unchanged). Since I is the kernel of the linear transformation from V onto F which sends a vector to the sum of its coefficients (called the augmentation map — cf. Section 7.3), I has dimension $n - 1$.

- (11) If $V = FG$ is the regular representation of G described in Example 2 above, then V has FG -submodules of dimensions 1 and $|G| - 1$ as in the preceding example:

$$N = \{\alpha_1 g_1 + \alpha_2 g_2 + \cdots + \alpha_n g_n \mid \alpha_1 = \alpha_2 = \cdots = \alpha_n\}$$

$$I = \{\alpha_1 g_1 + \alpha_2 g_2 + \cdots + \alpha_n g_n \mid \alpha_1 + \alpha_2 + \cdots + \alpha_n = 0\}.$$

In fact N and I are 2-sided ideals of FG (not just left ideals — note that N is in the center of FG). The ideal I is called the *augmentation ideal* of FG and N is called the *trace ideal* of FG .

Recall that in the study of a linear transformation T of a vector space V to itself we made V into an $F[x]$ -module (where x acted as T on V); our goal was to decompose V into a direct sum of cyclic submodules. In this way we were able to find a basis of V for which the matrix of T with respect to this basis was in some *canonical* form. Changing the basis of V did not change the module V but changed the matrix representation of T by similarity (i.e., changed the isomorphism between $GL(V)$ and $GL_n(F)$). We introduce the analogous terminology to describe when two FG -modules are the same up to a change of basis.

Definition. Two representations of G are *equivalent* (or *similar*) if the FG -modules affording them are isomorphic modules. Representations which are not equivalent are called *inequivalent*.

Suppose $\varphi : G \rightarrow GL(V)$ and $\psi : G \rightarrow GL(W)$ are equivalent representations (here V and W must be vector spaces over the same field F). Let $T : V \rightarrow W$ be

an FG -module isomorphism between them. Since T is, in particular, an F -module isomorphism, T is a vector space isomorphism, so V and W must have the same dimension. Furthermore, for all $g \in G$, $v \in V$ we have $T(g \cdot v) = g \cdot (T(v))$, since T is an isomorphism of FG -modules. By definition of the action of ring elements this means $T(\varphi(g)v) = \psi(g)(T(v))$, that is

$$T \circ \varphi(g) = \psi(g) \circ T \quad \text{for all } g \in G.$$

In particular, if we identify V and W as vector spaces, then two representations φ and ψ of G on a vector space V are equivalent if and only if there is some $T \in GL(V)$ such that $T \circ \varphi(g) \circ T^{-1} = \psi(g)$ for all $g \in G$. This T is a *simultaneous* change of basis for all $\varphi(g)$, $g \in G$.

In matrix terminology, two representations φ and ψ are equivalent if there is a fixed invertible matrix P such that

$$P\varphi(g)P^{-1} = \psi(g) \quad \text{for all } g \in G.$$

The linear transformation T or the matrix P above is said to *intertwine* the representations φ and ψ (it gives the “rule” for changing φ into ψ).

In order to study the decomposition of an FG -module into (direct sums of) submodules we shall need some terminology. We state these definitions for arbitrary rings since we shall be discussing direct sum decompositions in greater generality in the next section.

Definition. Let R be a ring and let M be a nonzero R -module.

- (1) The module M is said to be *irreducible* (or *simple*) if its only submodules are 0 and M ; otherwise M is called *reducible*.
- (2) The module M is said to be *indecomposable* if M cannot be written as $M_1 \oplus M_2$ for any nonzero submodules M_1 and M_2 ; otherwise M is called *decomposable*.
- (3) The module M is said to be *completely reducible* if it is a direct sum of irreducible submodules.
- (4) A representation is called *irreducible*, *reducible*, *indecomposable*, *decomposable* or *completely reducible* according to whether the FG -module affording it has the corresponding property.
- (5) If M is a completely reducible R -module, any direct summand of M is called a *constituent* of M (i.e., N is a constituent of M if there is a submodule N' of M such that $M = N \oplus N'$).

An irreducible module is, by definition, both indecomposable and completely reducible. We shall shortly give examples of indecomposable modules that are not irreducible.

If $R = FG$, an irreducible FG -module V is a nonzero F -vector space with no non-trivial, proper G -invariant subspaces. For example, if $\dim_F V = 1$ then V is necessarily irreducible (its only subspaces are 0 and V).

Suppose V is a finite dimensional FG -module and V is reducible. Let U be a G -invariant subspace. Form a basis of V by taking a basis of U and enlarging it to a

basis of V . Then for each $g \in G$ the matrix, $\varphi(g)$, of g acting on V with respect to this basis is of the form

$$\varphi(g) = \begin{pmatrix} \varphi_1(g) & \psi(g) \\ 0 & \varphi_2(g) \end{pmatrix}$$

where $\varphi_1 = \varphi|_U$ (with respect to the chosen basis of U) and φ_2 is the representation of G on V/U (and ψ is not necessarily a homomorphism — $\psi(g)$ need not be a square matrix). So reducible representations are those with a corresponding matrix representation whose matrices are in block upper triangular form.

Assume further that the FG -module V is decomposable, $V = U \oplus U'$. Take for a basis of V the union of a basis of U and a basis of U' . With this choice of basis the matrix for each $g \in G$ is of the form

$$\varphi(g) = \begin{pmatrix} \varphi_1(g) & 0 \\ 0 & \varphi_2(g) \end{pmatrix}$$

(i.e., $\psi(g) = 0$ for all $g \in G$). Thus decomposable representations are those with a corresponding matrix representation whose matrices are in block diagonal form.

Examples

- (1) As noted above, all degree 1 representations are irreducible, indecomposable and completely reducible. In particular, this applies to the trivial representation and to the representations described in Example 5 above.
- (2) If $|G| > 1$, the regular representation of G is reducible (the augmentation ideal and the trace ideal are proper nonzero submodules). We shall later determine the conditions under which this representation is completely reducible and how it decomposes into a direct sum.
- (3) For $n > 1$ the FS_n -module described in Example 10 above is reducible since N and I are proper, nonzero submodules. The module N is irreducible (being 1-dimensional) and if the characteristic of the field F does not divide n , then I is also irreducible.
- (4) The degree 2 representation of the dihedral group $D_{2n} = G$ described in Example 6 above is irreducible for $n \geq 3$. There are no G -invariant 1-dimensional subspaces since a rotation by $2\pi/n$ radians sends no line in $\mathbb{R}^2$ to itself. Similarly, the degree 2 complex representation of Q_8 described in Example 7 is irreducible since the given matrix $\varphi(i)$ has exactly two 1-dimensional eigenspaces (corresponding to its distinct eigenvalues $\pm\sqrt{-1}$) and these are not invariant under the matrix $\varphi(j)$. The degree 4 representation $\varphi : Q_8 \rightarrow GL_4(\mathbb{R})$ described in Example 8 can also be shown to be irreducible (see the exercises). We shall see, however, that if we view φ as a complex representation $\varphi : Q_8 \rightarrow GL_4(\mathbb{C})$ (just by considering the real entries of the matrices to be complex entries) then there is a *complex* matrix P such that $P^{-1}\varphi(g)P$ is a direct sum of 2×2 block matrices for all $g \in Q_8$. Thus an irreducible representation over a field F may become reducible when the field is extended.
- (5) Let $G = \langle g \rangle$ be cyclic of order n and assume F contains all the n^{th} roots of 1. As noted in Example 1 in the set of examples of group algebras, $F\langle g \rangle \cong F[x]/(x^n - 1)$. Thus the FG -modules are precisely the $F[x]$ -modules annihilated by $x^n - 1$. The latter (finite dimensional) modules are described, up to equivalence, by the Jordan Canonical Form Theorem.

If the minimal polynomial of g acting on an $F\langle g \rangle$ -module V has distinct roots in F , there is a basis of V such that g (hence all its powers) is represented by a diagonal

matrix (cf. Corollary 25, Section 12.3). In this case, V is a completely reducible $F\langle g \rangle$ -module (being a direct sum of 1-dimensional $\langle g \rangle$ -invariant subspaces). In general, the minimal polynomial of g acting on V divides $x^n - 1$ so if $x^n - 1$ has distinct roots in F , then V is a completely reducible $F\langle g \rangle$ -module. The polynomial $x^n - 1$ has distinct roots in F if and only if the characteristic of F does not divide n . This gives a sufficient condition for every $F\langle g \rangle$ -module to be completely reducible.

If the minimal polynomial of g acting on V does *not* have distinct roots (so the characteristic of F does divide n), the Jordan canonical form of g must have an elementary Jordan block of size > 1 . Since every linear transformation has a unique Jordan canonical form, g cannot be represented by a diagonal matrix, i.e., V is not completely reducible. It follows from results on cyclic modules in Section 12.3 that the (1-dimensional) eigenspace of g in any Jordan block of size > 1 admits no $\langle g \rangle$ -invariant complement, i.e., V is reducible but not completely reducible.

Specifically, let p be a prime, let $F = \mathbb{F}_p$ and let g be of order p . Let V be the 2-dimensional space over $\mathbb{F}_p$ with basis v, w and define an action of g on V by

$$g \cdot v = v \quad \text{and} \quad g \cdot w = v + w.$$

This endomorphism of V does have order p (in $GL(V)$) and the matrix of g with respect to this basis is the elementary Jordan block

$$\varphi(g) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Now V is reducible ($\text{span}\{v\}$ is a $\langle g \rangle$ -invariant subspace) but V is indecomposable (the above 2×2 elementary Jordan matrix is not similar to a diagonal matrix).

The first fundamental result in the representation theory of finite groups shows how Example 5 generalizes to noncyclic groups.

Theorem 1. (Maschke's Theorem) Let G be a finite group and let F be a field whose characteristic does not divide $|G|$. If V is any FG -module and U is any submodule of V , then V has a submodule W such that $V = U \oplus W$ (i.e., every submodule is a direct summand).

Remark: The hypothesis of Maschke's Theorem applies to any finite group when F has characteristic 0.

Proof: The idea of the proof of Maschke's Theorem is to produce an FG -module homomorphism

$$\pi : V \rightarrow U$$

which is a projection onto U , i.e., which satisfies the following two properties:

- (i) $\pi(u) = u$ for all $u \in U$
- (ii) $\pi(\pi(v)) = \pi(v)$ for all $v \in V$ (i.e., $\pi^2 = \pi$)

(in fact (ii) is implied by (i) and the fact that $\pi(V) \subseteq U$).

Suppose first that we can produce such an FG -module homomorphism and let $W = \ker \pi$. Since π is a module homomorphism, W is a submodule. We see that W is a direct sum complement to U as follows. If $v \in U \cap W$ then by (i), $v = \pi(v)$ whereas by definition of W , $\pi(v) = 0$. This shows $U \cap W = 0$. To show $V = U + W$ let v be

an arbitrary element of V and write $v = \pi(v) + (v - \pi(v))$. By definition, $\pi(v) \in U$. By property (ii) of π ,

$$\pi(v - \pi(v)) = \pi(v) - \pi(\pi(v)) = \pi(v) - \pi(v) = 0,$$

i.e., $v - \pi(v) \in W$. This shows $V = U + W$ and hence $V = U \oplus W$. To establish Maschke's Theorem it therefore suffices to find such an FG -module projection π .

Since U is a subspace it has a vector space direct sum complement W_0 in V (take a basis $\mathcal{B}_1$ of U , build it up to a basis $\mathcal{B}$ of V and let W_0 be the span of $\mathcal{B} - \mathcal{B}_1$). Thus $V = U \oplus W_0$ as vector spaces but W_0 need not be G -stable (i.e., need not be an FG -submodule). Let $\pi_0 : V \rightarrow U$ be the vector space projection of V onto U associated to this direct sum decomposition, i.e., π_0 is defined by

$$\pi_0(u + w) = u \quad \text{for all } u \in U, w \in W_0.$$

The key idea of the proof is to “average” π_0 over G to form an FG -module projection π . For each $g \in G$ define

$$g\pi_0g^{-1} : V \rightarrow U \quad \text{by} \quad g\pi_0g^{-1}(v) = g \cdot \pi_0(g^{-1} \cdot v), \quad \text{for all } v \in V$$

(here $\cdot$ denotes the action of elements of the ring FG). Since π_0 maps V into U and U is stable under the action of g we have that $g\pi_0g^{-1}$ maps V into U . Both g and g^{-1} act as F -linear transformations, so $g\pi_0g^{-1}$ is a linear transformation. Furthermore, if u is in the G -stable space U then so is $g^{-1}u$, and by definition of π_0 we have $\pi_0(g^{-1}u) = g^{-1}u$. From this we obtain that for all $g \in G$,

$$g\pi_0g^{-1}(u) = u \quad \text{for all } u \in U$$

(i.e., $g\pi_0g^{-1}$ is also a vector space projection of V onto U).

Let $n = |G|$ and view n as an element of F ($n = 1 + \cdots + 1$, n times). By hypothesis n is not zero in F and so has an inverse in F . Define

$$\pi = \frac{1}{n} \sum_{g \in G} g\pi_0g^{-1}.$$

Since π is a scalar multiple of a sum of linear transformations from V to U , it is also a linear transformation from V to U . Furthermore, each term in the sum defining π restricts to the identity map on the subspace U and so $\pi|_U$ is $1/n$ times the sum of n copies of the identity. These observations prove the following:

$\pi : V \rightarrow U$ is a linear transformation

$$\pi(u) = u \quad \text{for all } u \in U$$

$$\pi^2(v) = \pi(v) \quad \text{for all } v \in V.$$

It remains to show that π is an FG -module homomorphism (i.e., is FG -linear). It

suffices to prove that for all $h \in G$, $\pi(hv) = h\pi(v)$, for $v \in V$. In this case

$$\begin{aligned}\pi(hv) &= \frac{1}{n} \sum_{g \in G} g\pi_0(g^{-1}hv) \\ &= \frac{1}{n} \sum_{g \in G} h(h^{-1}g)\pi_0((g^{-1}h)v) \\ &= \frac{1}{n} \sum_{\substack{k=h^{-1}g \\ g \in G}} h(k\pi_0(k^{-1}v)) = h\pi(v)\end{aligned}$$

(as g runs over all elements of G , so does $k = h^{-1}g$ and the module element h may be brought outside the summation by the distributive law in modules). This establishes the existence of the FG -module projection π and so completes the proof.

The applications of Maschke's Theorem will be to finitely generated FG -modules. Unlike the situation of $F[x]$ -modules, however, finitely generated FG -modules are automatically finite dimensional vector spaces (the difference being that FG itself is finite dimensional, whereas $F[x]$ is not). Let V be an FG -module. If V is a finite dimensional vector space over F , then a fortiori V is finitely generated as an FG -module (any F basis gives a set of generators over FG). Conversely, if V is finitely generated as an FG -module, say by $v_1, \dots, v_k$, then one easily sees that V is spanned as a vector space by the finite set $\{g \cdot v_i \mid g \in G, 1 \leq i \leq k\}$. Thus

an FG -module is finitely generated if and only if it is finite dimensional.

Corollary 2. If G is a finite group and F is a field whose characteristic does not divide $|G|$, then every finitely generated FG -module is completely reducible (equivalently, every F -representation of G of finite degree is completely reducible).

Proof: Let V be a finitely generated FG -module. As noted above, V is finite dimensional over F , so we may proceed by induction on its dimension. If V is irreducible, it is completely reducible and the result holds. Suppose therefore that V has a proper, nonzero FG -submodule U . By Maschke's Theorem U has an FG -submodule complement W , i.e., $V = U \oplus W$. By induction, each of U and W are direct sums of irreducible submodules, hence so is V . This completes the induction.

Corollary 3. Let G be a finite group, let F be a field whose characteristic does not divide $|G|$ and let $\varphi : G \rightarrow GL(V)$ be a representation of G of finite degree. Then there is a basis of V such that for each $g \in G$ the matrix of $\varphi(g)$ with respect to this basis is block diagonal:

$$\begin{pmatrix} \varphi_1(g) & & & \\ & \varphi_2(g) & & \\ & & \ddots & \\ & & & \varphi_m(g) \end{pmatrix}$$

where φ_i is an irreducible matrix representation of G , $1 \leq i \leq m$.

Proof: By Corollary 2 we may write $V = U_1 \oplus U_2 \oplus \cdots \oplus U_m$, where U_i is an irreducible FG -submodule of V . Let $\mathcal{B}_i$ be a basis of U_i and let $\mathcal{B}$ be the union of the $\mathcal{B}_i$'s. For each $g \in G$, the matrix of $\varphi(g)$ with respect to the basis $\mathcal{B}$ is of the form in the corollary, where $\varphi_i(g)$ is the matrix of $\varphi(g)|_{U_i}$ with respect to the basis $\mathcal{B}_i$.

The converse of Maschke's Theorem is also true. Namely, if the characteristic of F does divide $|G|$, then G possesses (finitely generated) FG -modules which are not completely reducible. Specifically, the regular representation (i.e., the module FG itself) is not completely reducible.

In Section 18.2 we shall discuss the question of uniqueness of the constituents in direct sum decompositions of FG -modules into irreducible submodules.

EXERCISES

Let F be a field, let G be a finite group and let $n \in \mathbb{Z}^+$.

1. Prove that if $\varphi : G \rightarrow GL(V)$ is any representation, then φ gives a faithful representation of $G/\ker \varphi$.
2. Let $\varphi : G \rightarrow GL_n(F)$ be a matrix representation. Prove that the map $g \mapsto \det(\varphi(g))$ is a degree 1 representation.
3. Prove that the degree 1 representations of G are in bijective correspondence with the degree 1 representations of the abelian group G/G' (where G' is the commutator subgroup of G).
4. Let V be a (possibly infinite dimensional) FG -module (G is a finite group). Prove that for each $v \in V$ there is an FG -submodule containing v of dimension $\leq |G|$.
5. Prove that if $|G| > 1$ then every irreducible FG -module has dimension $< |G|$.
6. Write out the matrices $\varphi(g)$ for every $g \in G$ for each of the following representations that were described in the second set of examples:
 - (a) the representation of S_3 described in Example 3 (let $n = 3$ in that example)
 - (b) the representation of D_8 described in Example 6 (i.e., let $n = 4$ in that example and write out the values of all the sines and cosines, for all group elements)
 - (c) the representation of Q_8 described in Example 7
 - (d) the representation of Q_8 described in Example 8.
7. Let V be the 4-dimensional permutation module for S_4 described in Example 3 of the second set of examples. Let $\pi : D_8 \rightarrow S_4$ be the permutation representation of D_8 obtained from the action of D_8 by left multiplication on the set of left cosets of its subgroup $\langle s \rangle$. Make V into an FD_8 -module via π as described in Example 4 and write out the 4×4 matrices for r and s given by this representation with respect to the basis $e_1, \dots, e_4$.
8. Let V be the FS_n -module described in Examples 3 and 10 in the second set of examples.
 - (a) Prove that if v is any element of V such that $\sigma \cdot v = v$ for all $\sigma \in S_n$ then v is an F -multiple of $e_1 + e_2 + \cdots + e_n$.
 - (b) Prove that if $n \geq 3$, then V has a unique 1-dimensional submodule, namely the submodule N consisting of all F -multiples of $e_1 + e_2 + \cdots + e_n$.
9. Prove that the 4-dimensional representation of Q_8 on $\mathbb{H}$ described in Example 8 in the second set of examples is irreducible. [Show that any Q_8 -stable subspace is a left ideal.]
10. Prove that $GL_2(\mathbb{R})$ has no subgroup isomorphic to Q_8 . [This may be done by direct computation using generators and relations for Q_8 . Simplify these calculations by putting one generator in rational canonical form.]

11. Let $\varphi : S_n \rightarrow GL_n(F)$ be the matrix representation given by the permutation module described in Example 3 in the second set of examples, where the matrices are computed with respect to the basis $e_1, \dots, e_n$. Prove that $\det \varphi(\sigma) = \epsilon(\sigma)$ for all $\sigma \in S_n$, where $\epsilon(\sigma)$ is the sign of the permutation σ . [Check this on transpositions.]
12. Assume the characteristic of F is not 2. Let H be the set of $T \in M_n(F)$ such that T has exactly one nonzero entry in each row and each column and zeros elsewhere, and the nonzero entries are ± 1 . Prove that H is a subgroup of $GL_n(F)$ and that H is isomorphic to $E_{2^n} \rtimes S_n$ (semidirect product), where E_{2^n} is the elementary abelian group of order 2^n .

The next few exercises explore an important result known as Schur's Lemma and some of its consequences.

13. Let R be a ring and let M and N be simple (i.e., irreducible) R -modules.
 - (a) Prove that every nonzero R -module homomorphism from M to N is an isomorphism. [Consider its kernel and image.]
 - (b) Prove Schur's Lemma: if M is a simple R -module then $\text{Hom}_R(M, M)$ is a division ring (recall that $\text{Hom}_R(M, M)$ is the ring of all R -module homomorphisms from M to M , where multiplication in this ring is composition).
14. Let $\varphi : G \rightarrow GL(V)$ be a representation of G . The *centralizer* of φ is defined to be the set of all linear transformations, A , from V to itself such that $A\varphi(g) = \varphi(g)A$ for all $g \in G$ (i.e., the linear transformations of V which commute with all $\varphi(g)$'s).
 - (a) Prove that a linear transformation A from V to V is in the centralizer of φ if and only if it is an FG -module homomorphism from V to itself (so the centralizer of φ is the same as the ring $\text{Hom}_{FG}(V, V)$).
 - (b) Show that if z is in the center of G then $\varphi(z)$ is in the centralizer of φ .
 - (c) Assume φ is an irreducible representation (so V is a simple FG -module). Prove that if H is any finite abelian subgroup of $GL(V)$ such that $A\varphi(g) = \varphi(g)A$ for all $A \in H$ then H is cyclic (in other words, any finite abelian subgroup of the multiplicative group of units in the ring $\text{Hom}_{FG}(V, V)$ is cyclic). [By the preceding exercise, $\text{Hom}_{FG}(V, V)$ is a division ring, so this reduces to proving that a finite abelian subgroup of the multiplicative group of nonzero elements in a division ring is cyclic. Show that the division subring generated by an abelian subgroup of any division ring is a field and use Proposition 18, Section 9.5.]
 - (d) Show that if φ is a faithful irreducible representation then the center of G is cyclic.
 - (e) Deduce from (d) that if G is abelian and φ is any irreducible representation then $G/\ker \varphi$ is cyclic.
15. Exhibit all 1-dimensional complex representations of a finite cyclic group; make sure to decide which are inequivalent.
16. Exhibit all 1-dimensional complex representations of a finite abelian group. Deduce that the number of inequivalent degree 1 complex representations of a finite abelian group equals the order of the group. [First decompose the abelian group into a direct product of cyclic groups, then use the preceding exercise.]
17. Prove the following variant of Schur's Lemma for complex representations of abelian groups: if G is abelian, any irreducible complex representation, φ , of G is of degree 1 and $G/\ker \varphi$ is cyclic. [This can be done without recourse to Exercise 14 by using the observation that for any $g \in G$ the eigenspaces of $\varphi(g)$ are G -stable. Your proof that φ has degree 1 should also work for infinite abelian groups.]
18. Prove the following general form of Schur's Lemma for complex representations: if $\varphi : G \rightarrow GL_n(\mathbb{C})$ is an irreducible matrix representation and A is an $n \times n$ matrix com-

- muting with $\varphi(g)$ for all $g \in G$, then A is a scalar matrix. Deduce that if φ is a faithful, irreducible, complex representation then the center of G is cyclic and $\varphi(z)$ is a scalar matrix for all elements z in the center of G . [As in the preceding exercise, the eigenspaces of A are G -stable.]
19. Prove that if G is an abelian group then any finite dimensional complex representation of G is equivalent to a representation into diagonal matrices (i.e., any finite group of commuting matrices over $\mathbb{C}$ can be simultaneously diagonalized). [This can be done without recourse to Maschke's Theorem by looking at eigenspaces.]
 20. Prove that the number of degree 1 complex representations of any finite group G equals $|G : G'|$, where G' is the commutator subgroup of G . [Use Exercises 3 and 16.]
 21. Let G be a noncyclic abelian group acting by conjugation on an elementary abelian p -group V , where p is a prime not dividing the order of G .
 - (a) Prove that if W is an irreducible $\mathbb{F}_p G$ -submodule of V then there is some nonidentity element $g \in G$ such that $W \leq C_V(g)$ (here $C_V(g)$ is the subgroup of elements of V that are fixed by g under conjugation).
 - (b) Prove that V is generated by the subgroups $C_V(g)$ as g runs over all nonidentity elements of G .
 22. Let p be a prime, let P be a p -group and let F be a field of characteristic p . Prove that the only irreducible representation of P over F is the trivial representation. [Do this for a group of order p first using the fact that F contains all p^{th} roots of 1 (namely 1 itself). If P is not of order p , let z be an element of order p in the center of P , prove that z is in the kernel of the irreducible representation and apply induction to $P/\langle z \rangle$.]
 23. Let p be a prime, let P be a nontrivial p -group and let F be a field of characteristic p . Prove that the regular representation is not completely reducible. [Use the preceding exercise.]
 24. Let p be a prime, let P be a nontrivial p -group and let F be a field of characteristic p . Prove that the regular representation is indecomposable.

18.2 WEDDERBURN'S THEOREM AND SOME CONSEQUENCES

In this section we give a famous classification theorem due to Wedderburn which describes, in particular, the structure of the group algebra FG when the characteristic of F does not divide the order of G . From this classification theorem we shall derive various consequences, including the fact that for each finite group G there are only a finite number of nonisomorphic irreducible FG -modules. This result, together with Maschke's Theorem, in some sense completes the Hölder Program for representation theory of finite groups over such fields. The remainder of the book is concerned with developing techniques for determining and working with the irreducible representations as well as applying this knowledge to obtain group-theoretic information.

Theorem 4. (Wedderburn's Theorem) Let R be a nonzero ring with 1 (not necessarily commutative). Then the following are equivalent:

- (1) every R -module is projective
- (2) every R -module is injective
- (3) every R -module is completely reducible
- (4) the ring R considered as a left R -module is a direct sum:

$$R = L_1 \oplus L_2 \oplus \cdots \oplus L_n,$$

where each L_i is a simple module (i.e., a simple left ideal) with $L_i = Re_i$, for some $e_i \in R$ with

- (i) $e_i e_j = 0$ if $i \neq j$
 - (ii) $e_i^2 = e_i$ for all i
 - (iii) $\sum_{i=1}^n e_i = 1$
- (5) as rings, R is isomorphic to a direct product of matrix rings over division rings, i.e., $R = R_1 \times R_2 \times \cdots \times R_r$ where R_j is a two-sided ideal of R and R_j is isomorphic to the ring of all $n_j \times n_j$ matrices with entries in a division ring Δ_j , $j = 1, 2, \dots, r$. The integer r , the integers n_j , and the division rings Δ_j (up to isomorphism) are uniquely determined by R .

Proof: A proof of Wedderburn's Theorem is outlined in Exercises 1 to 10

Definition. A ring R satisfying any of the (equivalent) properties in Theorem 4 is called *semisimple with minimum condition*.

Rings R satisfying any of the equivalent conditions of Theorem 4 also satisfy the *minimum condition* or *descending chain condition (D.C.C)* on left ideals:

if $I_1 \supseteq I_2 \supseteq \cdots$ is a descending chain of left ideals of R
then there is an $N \in \mathbb{Z}^+$ such that $I_k = I_N$ for all $k \geq N$

(which explains the use of this term in the definition above). The rings we deal with will all have this minimum condition. For example, group algebras always have this property since in any strictly descending chain of ideals the vector space dimensions of the ideals (which are F -subspaces of FG) are strictly decreasing, hence the length of a strictly descending chain is at most the dimension of $FG (= |G|)$. We shall therefore use the term "semisimple" to mean "semisimple with minimum condition." The rings R_i in conclusion (5) of Wedderburn's Theorem are called the *Wedderburn components* of R and the direct product decomposition of R is called its *Wedderburn decomposition*. Note that Wedderburn's Theorem for commutative rings is a consequence of the classification of Artinian rings in Section 16.1. A commutative semisimple ring with minimum condition is an Artinian ring with Jacobson radical equal to zero and so is a direct product of fields (which are its Wedderburn components).

One should note that condition (5) is a two-sided condition which describes the overall structure of R completely (the ring operations in this direct product of rings are componentwise addition and multiplication). In particular it implies that a semisimple ring also has the minimum condition on right ideals. A useful way of thinking of the elements of the direct product $R_1 \times \cdots \times R_r$ in conclusion (5) is as $n \times n$ (block diagonal) matrices of the form

$$\begin{pmatrix} A_1 & & & \\ & A_2 & & \\ & & \ddots & \\ & & & A_r \end{pmatrix}$$

where A_i is an arbitrary $n_i \times n_i$ matrix with entries from Δ_i (here $n = \sum_{i=1}^r n_i$).

Recall from Section 10.5 that an R -module Q is *injective* if whenever Q is a submodule of any R -module M , then M has a submodule N such that $M = Q \oplus N$. Maschke's Theorem therefore implies:

Corollary 5. If G is a finite group and F is a field whose characteristic does not divide $|G|$, then the group algebra FG is a semisimple ring.

Before obtaining more precise information about how the invariants n , r , Δ_j , etc., relate to invariants in group rings FG for certain fields F , we first study the structure of matrix rings (i.e., the rings described in conclusions (4) and (5) of Wedderburn's Theorem). We introduce some terminology which is used extensively in ring theory. Recall that the *center* of the ring R is the subring of elements commuting with all elements in R ; it will be denoted by $Z(R)$ (the center will contain 1 if the ring has a 1).

Definition.

- (1) A nonzero element e in a ring R is called an *idempotent* if $e^2 = e$.
- (2) Idempotents e_1 and e_2 are said to be *orthogonal* if $e_1e_2 = e_2e_1 = 0$.
- (3) An idempotent e is said to be *primitive* if it cannot be written as a sum of two (commuting) orthogonal idempotents.
- (4) The idempotent e is called a *primitive central idempotent* if $e \in Z(R)$ and e cannot be written as a sum of two orthogonal idempotents in the ring $Z(R)$.

Proposition 6 describes the ideal structure of a matrix ring and Proposition 8 extends these results to direct products of matrix rings.

Proposition 6. Let Δ be a division ring, let $n \in \mathbb{Z}^+$, let R be the ring of all $n \times n$ matrices with entries from Δ and let I be the identity matrix (= the 1 of R).

- (1) The only two-sided ideals of R are 0 and R .
- (2) The center of R consists of the scalar matrices αI , where α is in the center of Δ : $Z(R) = \{\alpha I \mid \alpha \in Z(\Delta)\}$, and this is a field isomorphic to $Z(\Delta)$. In particular, if Δ is a field, the center of R is the subring of all scalar matrices. The only central idempotent in R is I (in particular, I is primitive).
- (3) Let e_i be the matrix with a 1 in position i , i and zeros elsewhere. Then $e_1, \dots, e_n$ are orthogonal primitive idempotents and $\sum_{i=1}^n e_i = I$.
- (4) $L_i = Re_i$ is the left ideal consisting of arbitrary entries in column i and zeros in all other columns. L_i is a simple left R -module. Every simple left R -module is isomorphic to L_1 (in particular, all L_i are isomorphic R -modules) and as a left R -module we have $R = L_1 \oplus \dots \oplus L_n$.

Before proving this proposition it will be useful to have the following result.

Lemma 7. Let R be an arbitrary nonzero ring.

- (1) If M and N are simple R -modules and $\varphi : M \rightarrow N$ is a nonzero R -module homomorphism, then φ is an isomorphism.
- (2) (*Schur's Lemma*) If M is a simple R -module, then $\text{Hom}_R(M, M)$ is a division ring.

Proof of Lemma 7: To prove (1) note that since φ is nonzero, $\ker \varphi$ is a proper submodule of M . By simplicity of M we have $\ker \varphi = 0$. Similarly, the image of φ is a nonzero submodule of the simple module N , hence $\varphi(M) = N$. This proves φ is bijective, so (1) holds.

By part (1), every nonzero element of the ring $\text{Hom}_R(M, M)$ is an isomorphism, hence has an inverse. This gives (2).

Proof of Proposition 6 Let A be an arbitrary matrix in R whose i, j entry is a_{ij} . Let E_{ij} be the matrix with a 1 in position i, j and zeros elsewhere. The following straightforward computations are left as exercises:

- (i) $E_{ij}A$ is the matrix whose i^{th} row equals the j^{th} row of A and all other rows are zero.
- (ii) AE_{ij} is the matrix whose j^{th} column equals the i^{th} column of A and all other columns are zero.
- (iii) $E_{pq}AE_{rs}$ is the matrix whose p, s entry is a_{qr} and all other entries are zero.

To prove (1) suppose J is any nonzero 2-sided ideal of R and let A be an element of J with a nonzero entry in position q, r . Given any $p, s \in \{1, \dots, n\}$ we obtain from (iii) that

$$E_{ps} = \frac{1}{a_{qr}} E_{pq} A E_{rs} \in J.$$

Since the Δ -linear combinations of $\{E_{ps} \mid 1 \leq p \leq n, 1 \leq s \leq n\}$ give all of R , it follows that $J = R$. This proves (1).

To prove (2) assume $A \in Z(R)$. Thus for all i, j we have $E_{ij}A = AE_{ij}$. From (i) and (ii) above it follows immediately that all off-diagonal entries of A are zero and all diagonal entries of A are equal. Thus $A = \alpha I$ for some $\alpha \in \Delta$. Furthermore, A must also commute with the set of all scalar matrices $\beta I, \beta \in \Delta$, i.e., α must commute with all elements of Δ . Finally, since $Z(R)$ is a field, it is immediate that it contains a unique idempotent (namely I). This establishes all parts of (2).

In part (3) it is clear that $e_1, \dots, e_n$ are orthogonal idempotents whose sum is I . We defer proving that they are primitive until we have established (4).

Next we prove (4). From (ii) above it follows that $Re_i = RE_{ii}$ is the set of matrices with arbitrary entries in the i^{th} column and zeros in all other columns. Furthermore, if A is any nonzero element of Re_i , then certainly $RA \subseteq Re_i$. The reverse inclusion holds because if a_{pi} is a nonzero entry of A , then by (i) above

$$e_i = E_{ii} = \frac{1}{a_{pi}} E_{ip} A \in RA.$$

This proves $Re_i = RA$ for any nonzero element $A \in Re_i$, and so Re_i must be a simple R -module.

Let M be any simple R -module. Since $Im = m$ for all $m \in M$ and since $I = \sum_{i=1}^n e_i$, there exists some i and some $m \in M$ such that $e_i m \neq 0$. For this i and m the map $re_i \mapsto re_i m$ is a nonzero R -module homomorphism from the simple R -module Re_i to the simple module M . By Lemma 7(1) it is an isomorphism. By (ii), the map $r \mapsto rE_{i1}$ gives $Re_i \cong Re_1$. Finally, every matrix is the direct sum of its columns so $R = L_1 \oplus \dots \oplus L_n$. This completes the proof of (4).

It remains to prove that the idempotents in part (3) are primitive. If $e_i = a + b$, for some orthogonal idempotents a and b , then we shall see that

$$L_i = Re_i = Ra \oplus Rb.$$

This will contradict the fact that L_i is a simple R -module. To establish the above direct sum note first that since $ab = ba = 0$, we have $ae_i = a \in Re_i$ and $be_i = b \in Re_i$. For all $r \in R$ we have $re_i = ra + rb$, hence $Re_i = Ra + Rb$. Moreover, $Ra \cap Rb = 0$ because if $ra = sb$ for some $r, s \in R$, then $ra = raa = sba = 0$ (recall $a = a^2$ and $ba = 0$). This completes all parts of the proof.

Proposition 8. Let $R = R_1 \times R_2 \times \cdots \times R_r$, where R_i is the ring of $n_i \times n_i$ matrices over the division ring Δ_i , for $i = 1, 2, \dots, r$.

- (1) Identify R_i with the i^{th} component of the direct product. Let z_i be the r -tuple with the identity of R_i in position i and zero in all other positions. Then $R_i = z_i R$ and for any $a \in R_i$, $z_i a = a$ and $z_j a = 0$ for all $j \neq i$. The elements $z_1, \dots, z_r$ are all of the primitive central idempotents of R . They are pairwise orthogonal and $\sum_{i=1}^r z_i = 1$.
- (2) Let N be any left R -module and let $z_i N = \{z_i x \mid x \in N\}$, $1 \leq i \leq r$. Then $z_i N$ is a left R -submodule of N , each $z_i N$ is an R_i -module on which R_j acts trivially for all $j \neq i$, and

$$N = z_1 N \oplus z_2 N \oplus \cdots \oplus z_r N.$$

- (3) The simple R -modules are the simple R_i -modules on which R_j acts trivially for $j \neq i$ in the following sense. Let M_i be the unique simple R_i -module (cf. Proposition 6). We may consider M_i as an R -module by letting R_j act trivially for all $j \neq i$. Then $M_1, \dots, M_r$ are pairwise nonisomorphic simple R -modules and any simple R -module is isomorphic to one of $M_1, \dots, M_r$. Explicitly, the R -module M_i is isomorphic to the simple left ideal $(0, \dots, 0, L^{(i)}, 0, \dots, 0)$ of all elements of R whose i^{th} component, $L^{(i)}$, consists of matrices with arbitrary entries in the first column and zeros elsewhere.
- (4) For any R -module N the R -submodule $z_i N$ is a direct sum of simple R -modules, each of which is isomorphic to the module M_i in (3). In particular, if M is a simple R -module, then there is a unique i such that $z_i M = M$ and for this index i we have $M \cong M_i$; for all $j \neq i$, $z_j M = 0$.
- (5) If each Δ_i equals the field F , then R is a vector space over F of dimension $\sum_{i=1}^r n_i^2$ and $\dim_F Z(R) = r$.

Proof: In part (1) since multiplication in the direct product of rings is componentwise it is clear that z_i times the element $(a_1, \dots, a_r)$ of R is the r -tuple with a_i in position i and zeros elsewhere. Thus $R_i = z_i R$, z_i is the identity in R_i and $z_i a = 0$ if $a \in R_j$ for any $j \neq i$. It is also clear that $z_1, \dots, z_r$ are pairwise orthogonal central idempotents whose sum is the identity of R . The central idempotents of R are, by definition, the idempotents in $Z(R) = F_1 \times F_2 \times \cdots \times F_r$, where F_i is the center of R_i . By Proposition 6, F_i is the field $Z(\Delta_i)$. If $w = (w_1, \dots, w_r)$ is any central idempotent then $w_i \in F_i$ for all i , and since $w^2 = w$ we have $w_i^2 = w_i$ in the field F_i . Since 0 and 1 are the only solutions to $x^2 = x$ in a field, the only central idempotents in R are r -tuples

whose entries are 0's and 1's. Thus $z_1, \dots, z_r$ are primitive central idempotents and since every central idempotent is a sum of these, they are the complete set of primitive central idempotents of R . This proves (1).

To prove (2) let N be any left R -module. First note that for any $z \in Z(R)$ the set $\{zx \mid x \in N\}$ is an R -submodule of N . In particular, $z_i N$ is an R -submodule. Let $z_i x \in z_i N$ and let $a \in R_j$ for some $j \neq i$. By (1) we have that $a = az_j$ and so $az_i x = (az_j)(z_i x) = az_j z_i x = 0$ because $z_i z_j = 0$. Thus the R -submodule $z_i N$ is acted on trivially by R_j for all $j \neq i$. For each $x \in N$ we have by (1) that $x = 1x = z_1 x + \dots + z_r x$, hence $N = z_1 N + \dots + z_r N$. Finally, this sum is direct because if, for instance, $x \in z_1 N \cap (z_2 N + \dots + z_r N)$, then $x = z_1 x$ whereas z_1 times any element of $z_2 N + \dots + z_r N$ is zero. This proves (2).

In part (3) first note that an R_i -module M becomes an R -module when R_j is defined to act trivially on M for all $j \neq i$. For such a module M the R -submodules are the same as the R_i -submodules. Thus M_i is a simple R -module for each i since it is a simple R_i -module.

Next, let M be a simple R -module. By (2), $M = z_1 M \oplus \dots \oplus z_r M$. Since M has no nontrivial proper R -submodules, there must be a unique i such that $M = z_i M$ and $z_j M = 0$ for all $j \neq i$. Thus the simple R -module M is annihilated by R_j for all $j \neq i$. This implies that the R -submodules of M are the same as the R_i -submodules of M , so M is therefore a simple R_i -module. By Proposition 6, M is isomorphic as an R_i -module to M_i . Since R_j acts trivially on both M and M_i for all $j \neq i$, it follows that the R_i -module isomorphism may be viewed as an R -module isomorphism as well.

Suppose $i \neq j$ and suppose $\varphi : M_i \rightarrow M_j$ is an R -module isomorphism. If $s_i \in M_i$ then $s_i = z_i s_i$ so

$$\varphi(s_i) = \varphi(z_i s_i) = z_i \varphi(s_i) = 0,$$

since $\varphi(s_i) \in M_j$ and z_i acts trivially on M_j . This contradicts the fact that φ is an isomorphism and proves that $M_1, \dots, M_r$ are pairwise nonisomorphic simple R -modules.

Finally, the left ideal of R described in (3) is acted on trivially by R_j for all $j \neq i$ and, by Proposition 6, it is up to isomorphism the unique simple R_i -module. This left ideal is therefore a simple R -module which is isomorphic to M_i . This proves (3).

For part (4) we have already proved that if M is any simple R -module then there is a unique i such that $z_i M = M$ and $z_j M = 0$ for all $j \neq i$. Furthermore, we have shown that for this index i the simple R -module M is isomorphic to M_i . Now let N be any R -module. Then $z_i N$ is a module over R_i which is acted on trivially by R_j for all $j \neq i$. By Wedderburn's Theorem $z_i N$ is a direct sum of simple R -modules. Since each of these simple summands is acted on trivially by R_j for all $j \neq i$, each is isomorphic to M_i . This proves (4).

In part (5) if each Δ_i equals the field F , then as an F -vector space

$$R \cong M_{n_1}(F) \oplus M_{n_2}(F) \oplus \dots \oplus M_{n_r}(F).$$

Each matrix ring $M_{n_i}(F)$ has dimension n_i^2 over F , hence R has dimension $\sum_{i=1}^r n_i^2$ over F . Furthermore, the center of each $M_{n_i}(F)$ is 1-dimensional (since by Proposition 6(2) it is isomorphic to F), hence $Z(R)$ has dimension r over F . This completes the proof of the proposition.

We now apply Wedderburn's Theorem (and the above ring-theoretic calculations) to the group algebra FG . First of all, in order to apply Wedderburn's Theorem we need the characteristic of F not to divide $|G|$. In fact, since we shall be dealing with numerical data in the sections to come it will be convenient to have the characteristic of F equal to 0. Secondly, it will simplify matters if we force all the division rings which will appear in the Wedderburn decomposition of FG to equal the field F — we shall prove that imposing the condition that F be algebraically closed is sufficient to ensure this. To simplify notation we shall therefore take $F = \mathbb{C}$ for most of the remainder of the text. The reader can easily check that any algebraically closed field of characteristic 0 (e.g., the field of all algebraic numbers) can be used throughout in place of $\mathbb{C}$.

By Corollary 5 the ring $\mathbb{C}G$ is semisimple so by Wedderburn's Theorem

$$\mathbb{C}G \cong R_1 \times R_2 \times \cdots \times R_r$$

where R_i is the ring of $n_i \times n_i$ matrices over some division ring Δ_i . Thinking of the elements of this direct product as $n \times n$ block matrices ($n = \sum_{i=1}^r n_i$) where the i^{th} block has entries from Δ_i , the field $\mathbb{C}$ appears in this direct product as scalar matrices and is contained in the center of $\mathbb{C}G$. Note that each Δ_i is a vector space over $\mathbb{C}$ of dimension $\leq n$. The next result shows that this implies each $\Delta_i = \mathbb{C}$.

Proposition 9. If Δ is a division ring that is a finite dimensional vector space over an algebraically closed field F and $F \subseteq Z(\Delta)$, then $\Delta = F$.

Proof: Since $F \subseteq Z(\Delta)$, for each $\alpha \in \Delta$ the division ring generated by α and F is a field. Also, since Δ is finite dimensional over F the field $F(\alpha)$ is a finite extension of F . Because F is algebraically closed it has no nontrivial finite extensions, hence $F(\alpha) = F$ for all $\alpha \in \Delta$, i.e., $\Delta = F$.

This proposition proves that each R_i in the Wedderburn decomposition of $\mathbb{C}G$ is a matrix ring over $\mathbb{C}$:

$$R_i = M_{n_i}(\mathbb{C}).$$

Now Proposition 8(5) implies that

$$\sum_{i=1}^r n_i^2 = |G|.$$

The final application in this section is to prove that r (= the number of Wedderburn components in $\mathbb{C}G$) equals the number of conjugacy classes of G . To see this, first note that Proposition 8(5) asserts that $r = \dim_{\mathbb{C}} Z(\mathbb{C}G)$. We compute this dimension in another way.

Let $\mathcal{K}_1, \dots, \mathcal{K}_s$ be the distinct conjugacy classes of G (recall that these partition G). For each conjugacy class $\mathcal{K}_i$ of G let

$$X_i = \sum_{g \in \mathcal{K}_i} g \in \mathbb{C}G.$$

Note that X_i and X_j have no common terms for $i \neq j$, hence they are linearly independent elements of $\mathbb{C}G$. Furthermore, since conjugation by a group element permutes the

elements of each class, $h^{-1}X_i h = X_i$, i.e., X_i commutes with all group elements. This proves that $X_i \in Z(\mathbb{C}G)$.

We show the X_i 's form a basis of $Z(\mathbb{C}G)$, which will prove $s = \dim_{\mathbb{C}} Z(\mathbb{C}G) = r$. Since the X_i 's are linearly independent it remains to show they span $Z(\mathbb{C}G)$. Let $X = \sum_{g \in G} \alpha_g g$ be an arbitrary element of $Z(\mathbb{C}G)$. Since $h^{-1}Xh = X$,

$$\sum_{g \in G} \alpha_g h^{-1}gh = \sum_{g \in G} \alpha_g g.$$

Since the elements of G form a basis of $\mathbb{C}G$ the coefficients of g in the above two sums are equal:

$$\alpha_{hgh^{-1}} = \alpha_g.$$

Since h was arbitrary, every element in the same conjugacy class of a fixed group element g has the same coefficient in X , hence X can be written as a linear combination of the X_i 's.

We summarize these results in the following theorem.

Theorem 10. Let G be a finite group.

- (1) $\mathbb{C}G \cong M_{n_1}(\mathbb{C}) \times M_{n_2}(\mathbb{C}) \times \cdots \times M_{n_r}(\mathbb{C})$.
- (2) $\mathbb{C}G$ has exactly r distinct isomorphism types of irreducible modules and these have complex dimensions $n_1, n_2, \dots, n_r$ (and so G has exactly r inequivalent irreducible complex representations of the corresponding degrees).
- (3) $\sum_{i=1}^r n_i^2 = |G|$.
- (4) r equals the number of conjugacy classes in G .

Corollary 11.

- (1) Let A be a finite abelian group. Every irreducible complex representation of A is 1-dimensional (i.e., is a homomorphism from A into $\mathbb{C}^\times$) and A has $|A|$ inequivalent irreducible complex representations. Furthermore, every finite dimensional complex matrix representation of A is equivalent to a representation into a group of diagonal matrices.
- (2) The number of inequivalent (irreducible) degree 1 complex representations of any finite group G equals $|G/G'|$.

Proof: If A is abelian, $\mathbb{C}A$ is a commutative ring. Since a $k \times k$ matrix ring is not commutative whenever $k > 1$ we must have each $n_i = 1$. Thus $r = |A|$ (= the number of conjugacy classes of A). Since every $\mathbb{C}A$ -module is a direct sum of irreducible submodules, there is a basis such that the matrices are diagonal with respect to this basis. This establishes the first part of the corollary.

For a general group G , every degree 1 representation, φ , is a homomorphism of G into $\mathbb{C}^\times$. Thus φ factors through G/G' . Conversely, every degree 1 representation of G/G' gives, by composition with the natural projection $G \rightarrow G/G'$, a degree 1 representation of G . The degree 1 representations of G are therefore precisely the irreducible representations of the abelian group G/G' . Part (2) is now immediate from (1).

Examples

- (1) The irreducible complex representations of a finite abelian group A (i.e., the homomorphisms from A into $\mathbb{C}^\times$) can be explicitly described as follows: decompose A into a direct product of cyclic groups

$$A \cong C_1 \times \cdots \times C_n$$

where $|C_i| = |\langle x_i \rangle| = d_i$. Map each x_i to a (not necessarily primitive) d_i^{th} root of 1 and extend this to all powers of x_i to give a homomorphism. Since there are d_i choices for the image of each x_i , the number of distinct homomorphisms of A into $\mathbb{C}^\times = GL_1(\mathbb{C})$ defined by this process equals $|A|$. By Corollary 11, these are all the irreducible representations of A . Note that it is necessary that the field contain the appropriate roots of 1 in order to realize these representations. An exercise below explores the irreducible representations of cyclic groups over $\mathbb{Q}$.

- (2) Let $G = S_3$. By Theorem 10 the number of irreducible complex representations of G is three (= the number of conjugacy classes of S_3). Since the sum of the squares of the degrees is 6, the degrees must be 1, 1 and 2. The two degree 1 representations are immediately evident: the trivial representation and the representation of S_3 into $\{\pm 1\}$ given by mapping a permutation to its sign (i.e., $\sigma \mapsto +1$ if σ is an even permutation and $\sigma \mapsto -1$ if σ is an odd permutation). The degree 2 representation can be found by decomposing the permutation representation on 3 basis vectors (described in Section 1) into irreducibles as follows: let S_3 act on the basis vectors e_1, e_2, e_3 of a vector space V by permuting their indices. The vector $t = e_1 + e_2 + e_3$ is a nonzero fixed vector, so t spans a 1-dimensional G -invariant subspace (which is a copy of the trivial representation). By Maschke's Theorem there is a 2-dimensional G -invariant complement, I . Note that the permutation representation is not a sum of degree 1 representations: otherwise it could be represented by diagonal matrices and the permutations would commute in their action — this is impossible since the representation is faithful and G is non-abelian. Thus I cannot be decomposed further, so I affords *the* irreducible 2-dimensional representation. Indeed, I is the “augmentation” submodule described in Section 1:

$$I = \{w \in V \mid w = \alpha_1 e_1 + \alpha_2 e_2 + \alpha_3 e_3 \text{ with } \alpha_1 + \alpha_2 + \alpha_3 = 0\}.$$

Clearly $e_1 - e_2$ and $e_2 - e_3$ are independent vectors in I , hence they form a basis for this 2-dimensional space. With respect to this basis of I we obtain a matrix representation of S_3 and, for example, this matrix representation on two elements of S_3 is

$$(1\ 2) \mapsto \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad (1\ 2\ 3) \mapsto \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}.$$

- (3) We decompose the regular representation over $\mathbb{C}$ of an arbitrary finite group. Recall that this is the representation afforded by the left $\mathbb{C}G$ -module $\mathbb{C}G$ itself. By Theorem 10, $\mathbb{C}G$ is first of all a direct product of two-sided ideals:

$$\mathbb{C}G \cong M_{n_1}(\mathbb{C}) \times M_{n_2}(\mathbb{C}) \times \cdots \times M_{n_r}(\mathbb{C}).$$

Now by Proposition 6(4) each $M_{n_i}(\mathbb{C})$ decomposes further as a direct sum of n_i isomorphic simple left ideals. These left ideals give a complete set of isomorphism classes of irreducible $\mathbb{C}G$ -modules. Thus the regular representation (over $\mathbb{C}$) of G decomposes as *the direct sum of all irreducible representations of G , each appearing with multiplicity equal to the degree of that irreducible representation.*

We record one additional property of $\mathbb{C}G$ which we shall prove in Section 19.2.

Theorem 12. The degree of each complex irreducible representation of a finite group G divides the order of G , i.e., in the notation of Theorem 10, each n_i divides $|G|$ for $i = 1, 2, \dots, r$.

In the next section we shall describe the primitive central idempotents of $\mathbb{C}G$ in terms of the group elements.

EXERCISES

Let G be a finite group and let R be a ring with 1.

1. Prove that conditions (1) and (2) of Wedderburn's Theorem are equivalent.
2. Prove that (3) implies (2) in Wedderburn's Theorem. [Let Q be a submodule of an R -module N . Use Zorn's Lemma to show there is a submodule M maximal with respect to $Q \cap M = 0$. If $Q + M = N$, then (2) holds; otherwise let M_1 be the complete preimage in N of some simple module in N/M not contained in $(Q + M)/M$, and argue that M_1 contradicts the maximality of M .]
3. Prove that (4) implies (3) in Wedderburn's Theorem. [Let N be a nonzero R -module. First show N contains simple submodules by considering a cyclic submodule. Then use Zorn's Lemma applied to the set of direct sums of simple submodules (appropriately ordered) to show that N contains a maximal completely reducible submodule M . If $M \neq N$ let M_1 be the complete preimage in N of a simple module in N/M and contradict the maximality of M .]
4. Prove that (5) implies (4) in Wedderburn's Theorem. [Use the methods in the proofs of Propositions 6 and 8 to decompose each R_i as a left R -module.]

The next six exercises establish some general results about rings and modules that imply the remaining implication of Wedderburn's Theorem: (2) implies (5). In these exercises assume R satisfies (2): every R -module is injective.

5. Show that R has the descending chain condition (D.C.C.) on left ideals. Deduce that R is a finite direct sum of left ideals. [If not, then show that as a left R -module R is a direct sum of an infinite number of nonzero submodules. Derive a contradiction by writing the element 1 in this direct sum.]
6. Show that $R = R_1 \times R_2 \times \cdots \times R_r$ where R_i is a 2-sided ideal and a simple ring (i.e., has no proper, nonzero 2-sided ideals). Show each R_i has an identity and satisfies D.C.C. on left ideals. [Use the preceding exercise to show R has a minimal 2-sided ideal R_1 . As a left R -module $R = R_1 \oplus R'$ for some left ideal R' . Show R' is a right ideal and proceed inductively using D.C.C.]
7. Let S be a simple ring with 1 satisfying D.C.C. on left ideals and let L be a minimal left ideal in S . Show that $S \cong L^n$ as left S -modules, where $L^n = L \oplus \cdots \oplus L$ with n factors. [Argue by simplicity that $LS = S$ so $1 = l_1 s_1 + \cdots + l_n s_n$ for some $l_i \in L$ and $s_i \in S$ with n minimal. Show that the map $(x_1, \dots, x_n) \mapsto x_1 s_1 + \cdots + x_n s_n$ is a surjective homomorphism of left S -modules; use the minimality of L and n to show it is an injection.]
8. Let A be any ring with 1, let L be any left A -module and let L^n be the direct sum of n copies of L with itself.
 - (a) Prove the ring isomorphism $\text{Hom}_A(L^n, L^n) \cong M_n(D)$, where $D = \text{Hom}_A(L, L)$ (multiplication in the ring $\text{Hom}_A(X, X)$ is function composition, cf. Proposition 2(4) in Section 10.2).

- (b) Deduce that if L is a simple A -module, then $\text{Hom}_A(L^n, L^n)$ is isomorphic to a matrix ring over a division ring. [Use Schur's Lemma and (a).]
- (c) Prove the ring isomorphism $\text{Hom}_A(A, A) \cong A^{\text{opp}}$, where A^{opp} is the opposite ring to A (the elements and addition are the same as in A but the value of the product $x \cdot y$ in A^{opp} is yx , computed in A), cf. the end of Section 17.4. [Any homomorphism is determined by its value on 1.]
9. Prove that if S is a simple ring with 1 satisfying D.C.C. on left ideals then $S \cong M_n(\Delta)$ for some division ring Δ . (This result together with Exercise 6 completes the existence part of the proof that (2) implies (5) in Wedderburn's Theorem). [Use Exercises 7 and 8 to show $S^{\text{opp}} \cong \text{Hom}_S(L^n, L^n) \cong M_n(D)$ for some division ring D . Then show $S \cong M_n(\Delta)$, where Δ is the division ring D^{opp} .]
10. Prove that Δ and n in the isomorphism $S \cong M_n(\Delta)$ of the previous exercise are uniquely determined by S (proving the uniqueness statement in Wedderburn's Theorem), as follows. Suppose $S = M_n(\Delta) \cong M_{n'}(\Delta')$ as rings, where Δ and Δ' are division rings.
- (a) Prove that $\Delta \cong \text{Hom}_S(L, L)$ where L is a minimal left ideal in S . Deduce that $\Delta \cong \Delta'$. [Use Proposition 6(4).]
- (b) Prove that a finitely generated (left) module over a division ring Δ has a "basis" (a linearly independent generating set), and that any two bases have the same cardinality. Deduce that $n = n'$. [Mimic the proof of Corollary 4(2) of Section 11.1.]
11. Prove that if R is a ring with 1 such that every R -module is free then R is a division ring.
12. Let F be a field, let $f(x) \in F[x]$ and let $R = F[x]/(f(x))$. Find necessary and sufficient conditions on the factorization of $f(x)$ in $F[x]$ so that R is a semisimple ring. When R is semisimple, describe its Wedderburn decomposition. [See Proposition 16 in Section 9.5.]
13. Let G be the cyclic group of order n and let $R = \mathbb{Q}G$. Describe the Wedderburn decomposition of R and find the number and the degrees of the irreducible representations of G over $\mathbb{Q}$. In particular, show that if $n = p$ is a prime then G has exactly one nontrivial irreducible representation over $\mathbb{Q}$ and this representation has degree $p - 1$. [Recall from the first example in Section 1 that $\mathbb{Q}G = \mathbb{Q}[x]/(x^n - 1)$. Use Proposition 16 in Section 9.5 and results from Section 13.6.]
14. Let p be a prime and let $F = \mathbb{F}_p$ be the field of order p . Let G be the cyclic group of order 3 and let $R = FG$. For each of $p = 2$ and $p = 7$ describe the Wedderburn decomposition of R and find the number and the degrees of the irreducible representations of G over F .
15. Prove that if P is a p -group for some prime p , then P has a faithful irreducible complex representation if and only if $Z(P)$ is cyclic. [Use Exercise 18 in Section 1, Theorem 6.1(2) and Example 3.]
16. Prove that if V is an irreducible FG -module and F is an algebraically closed field then $\text{Hom}_{FG}(V, V)$ is isomorphic to F (as a ring).
17. Let F be a field, let $R = M_n(F)$ and let M be the unique irreducible R -module. Prove that $\text{Hom}_R(M, M)$ is isomorphic to F (as a ring).
18. Find all 2-sided ideals of $M_n(\mathbb{Z})$.

18.3 CHARACTER THEORY AND THE ORTHOGONALITY RELATIONS

In general, for groups of large order the representations are difficult to compute and unwieldy if not impossible to write down. For example, a matrix representation of degree 100 involves matrices with 10,000 entries, and a number of 100×100 matrices

may be required to describe the representation, even on a set of generators for the group. There are, however, some striking examples where large degree representations have been computed and used effectively. One instance of this is a construction of the simple group J_1 by Z. Janko in 1965 (the existence problem for simple groups was discussed at the end of Section 6.2). Janko was investigating certain properties of simple groups and he found that if any simple group possessed these properties, then it would necessarily have order 175,560 and would be generated by two elements. Furthermore, he proved that a hypothetical simple group with these properties must have a 7-dimensional representation over the field $\mathbb{F}_{11}$ with two generators mapping to the two matrices

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \text{ and } \begin{pmatrix} -3 & 2 & -1 & -1 & -3 & -1 & -3 \\ -2 & 1 & 1 & 3 & 1 & 3 & 3 \\ -1 & -1 & -3 & -1 & -3 & -3 & 2 \\ -1 & -3 & -1 & -3 & -3 & 2 & -1 \\ -3 & -1 & -3 & -3 & 2 & -1 & -1 \\ 1 & 3 & 3 & -2 & 1 & 1 & 3 \\ 3 & 3 & -2 & 1 & 1 & 3 & 1 \end{pmatrix}$$

(note that for any simple group S , every representation of S into $GL_n(F)$ which does not map all group elements to the identity matrix is a faithful representation, so S is isomorphic to its image in $GL_n(F)$). In particular, Janko's calculations showed that the simple group satisfying his properties was unique, if it existed. M. Ward was able to show that these two matrices do generate a subgroup of $GL_7(\mathbb{F}_{11})$ of order 175,560 and it follows that there does exist a simple group satisfying Janko's properties.

In a similar vein, S. Norton, R. Parker and J. Thackray constructed the simple group J_4 of order 86,775,571,046,077,562,880 using a 112-dimensional representation over $\mathbb{F}_2$. This group was shown to be generated by two elements, and explicit matrices in $GL_{112}(\mathbb{F}_2)$ for these two generators were computed in the course of their analysis.

In 1981, R. Griess constructed the largest of the sporadic groups, the so called *Monster*, of order

$$2^{46} \cdot 3^{20} \cdot 5^9 \cdot 7^6 \cdot 11^2 \cdot 13^3 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71.$$

His proof involves calculations of automorphisms of an algebra over $\mathbb{C}$ of dimension 196,884 and leads to a construction of the Monster by means of a representation of this degree.

By analogy, in general it is difficult to write out the explicit permutations associated to a permutation representation $\varphi : G \rightarrow S_n$ for large degrees n . There are, however, numerical invariants such as the signs and the cycle types of the permutations $\pi(g)$ and these numerical invariants might be easier to compute than the permutations themselves (i.e., it may be possible to determine the cycle types of elements without actually having to write out the permutations themselves, as in the computation of Galois groups over $\mathbb{Q}$ in Section 14.8). These invariants alone may provide enough information in a given situation to carry out some analysis, such as prove that a given group is not simple (as illustrated in Section 6.2). Furthermore, the invariants just mentioned do not depend on the labelling of the set $\{1, 2, \dots, n\}$ (i.e., they are independent of a "change of basis" in S_n) and they are the same for elements that are conjugate in G .

In this section we show how to attach numerical invariants to linear representations. These invariants depend only on the equivalence class (isomorphism type) of the representation. In other words, for each representation $\varphi : G \rightarrow GL_n(F)$ we shall attach an element of F to each matrix $\varphi(g)$ and we shall see that this number can, in many instances, be computed without knowing the matrix $\varphi(g)$. Moreover, we shall see that these invariants are independent of the similarity class of φ (i.e., are the same for a fixed $g \in G$ if the representation φ is replaced by an equivalent representation) and that they, in some sense, characterize the similarity classes of representations of G .

Throughout this section G is a finite group and, for the moment, F is an arbitrary field. All representations considered are assumed to be finite dimensional.

Definition.

- (1) A *class function* is any function from G into F which is constant on the conjugacy classes of G , i.e., $f : G \rightarrow F$ such that $f(g^{-1}xg) = f(x)$ for all $g, x \in G$.
- (2) If φ is a representation of G afforded by the FG -module V , the *character* of φ is the function

$$\chi : G \rightarrow F \quad \text{defined by} \quad \chi(g) = \text{tr } \varphi(g),$$

where $\text{tr } \varphi(g)$ is the trace of the matrix of $\varphi(g)$ with respect to some basis of V (i.e., the sum of the diagonal entries of that matrix). The character is called *irreducible* or *reducible* according to whether the representation is irreducible or reducible, respectively. The *degree* of a character is the degree of any representation affording it.

In the notation of the second part of this definition we shall also refer to χ as the character afforded by the FG -module V . In general, a character is *not* a homomorphism from a group into either the additive or multiplicative group of the field.

Examples

- (1) The character of the trivial representation is the function $\chi(g) = 1$ for all $g \in G$. This character is called the *principal* character of G .
- (2) For degree 1 representations, the character and the representation are usually identified (by identifying a 1×1 matrix with its entry). Thus for abelian groups, irreducible complex representations and their characters are the same (cf. Corollary 11).
- (3) Let $\Pi : G \rightarrow S_n$ be a permutation representation and let φ be the resulting linear representation on the basis $e_1, \dots, e_n$ of the vector space V :

$$\varphi(g)(e_i) = e_{\Pi(g)(i)}$$

(cf. Example 4 of Section 1). With respect to this basis the matrix of $\varphi(g)$ has a 1 in the diagonal entry i, i if $\Pi(g)$ fixes i ; otherwise, the matrix of $\varphi(g)$ has a zero in position i, i . Thus if π is the character of φ then

$$\pi(g) = \text{the number of fixed points of } g \text{ on } \{1, 2, \dots, n\}.$$

In particular, if Π is the permutation representation obtained from left multiplication on the set of left cosets of some subgroup H of G then the resulting character is called the *permutation character* of G on H .